

Module 7

Power Electronic Converters (Chap. 12-13 of Krause's Book)

Most Important Topics and Concepts

- Full-bridge square-wave inverters
- Power quality and harmonic control
- Pulse Width Modulation (PWM)
- Three-phase six-step and PWM-controlled inverters
- Multi-level inverters
- Space Vector Modulation (SVM)
- Average-Value Models (AVMs)

$ac \rightarrow dc$: rectifier
 $dc \rightarrow ac$: inverter

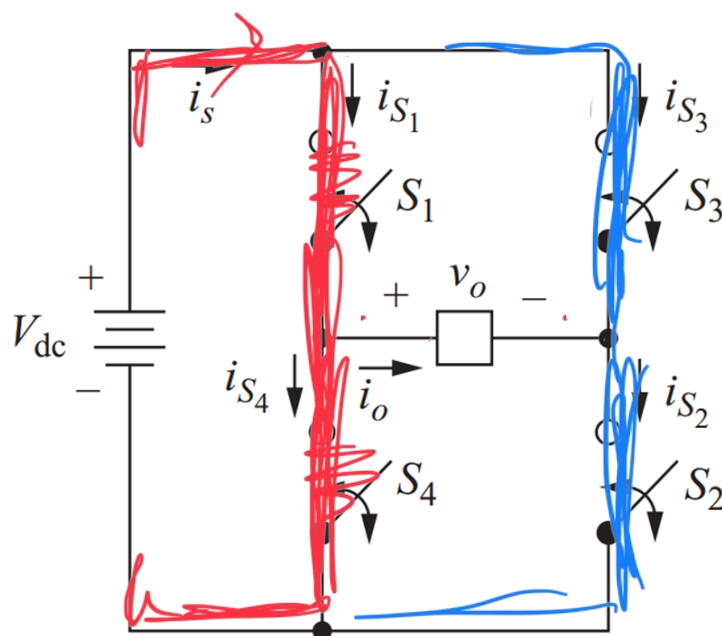


Full-Bridge Inverter

MosFets
10 BTs
100 Ts

❑ **Objective:** Producing an AC output for the load from a DC source

❖ Synthesizing an ac output from a dc input:



Output Voltage

Switching Options

S1 & S2 ON:

$$v_o = +V_{dc}$$

S3 & S4 ON:

$$v_o = -V_{dc}$$

S1 & S3 ON:

$$v_o = 0$$

S2 & S4 ON:

$$v_o = 0$$

S1 & S4 ON:

S2 & S3 ON:

- Turning ON both of the switches in the same leg short circuits the source! (aka shoot-through fault)

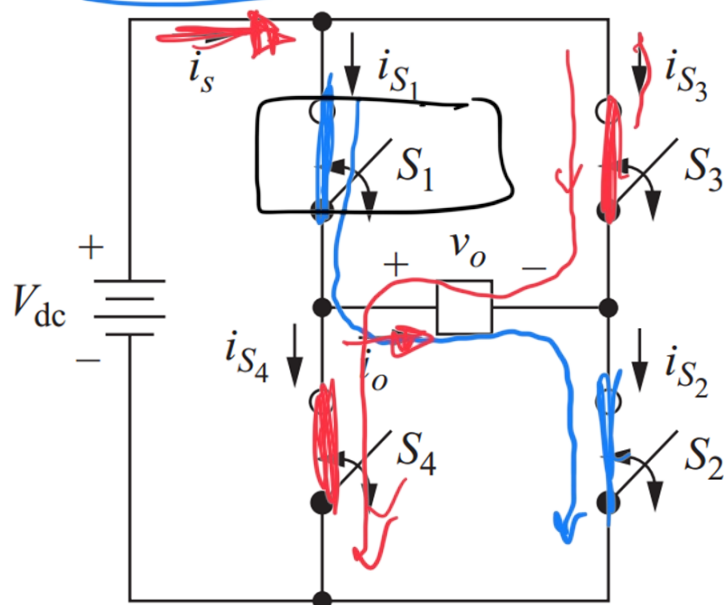


- The switching logic should include blanking time or dead-band to prevent both switches close simultaneously

Square-Wave Inverter

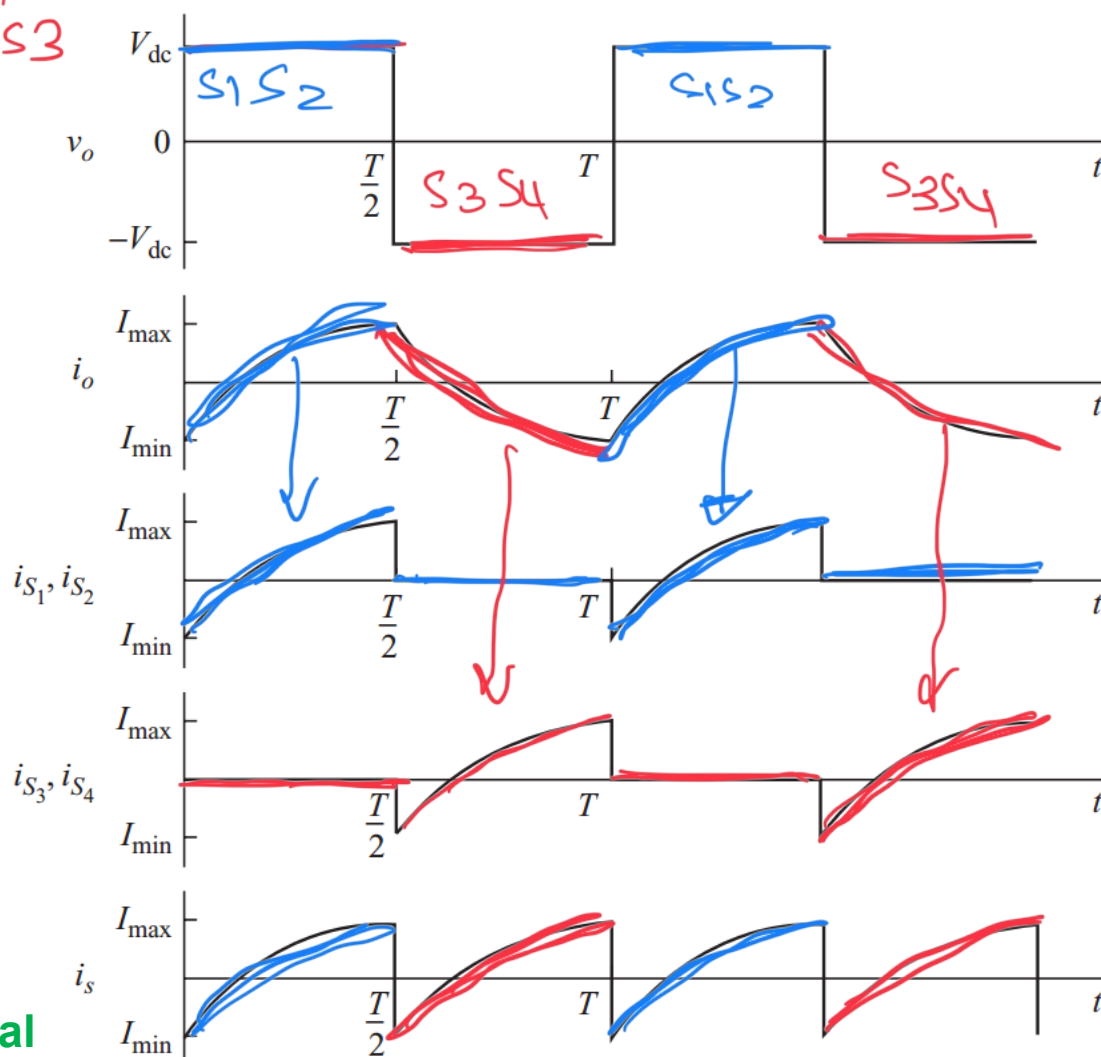
❑ **Objective:** Producing a square-wave output voltage from a DC source

❖ **RL Load:** $i_s = i_{s1} + i_{s3}$



S1 & S2 ON: $v_o = V_{dc}$

S3 & S4 ON: $v_o = -V_{dc}$



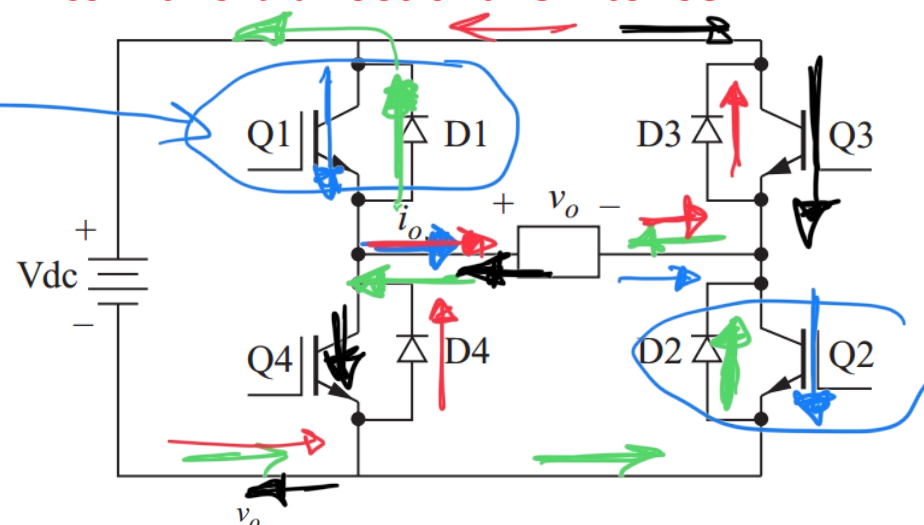
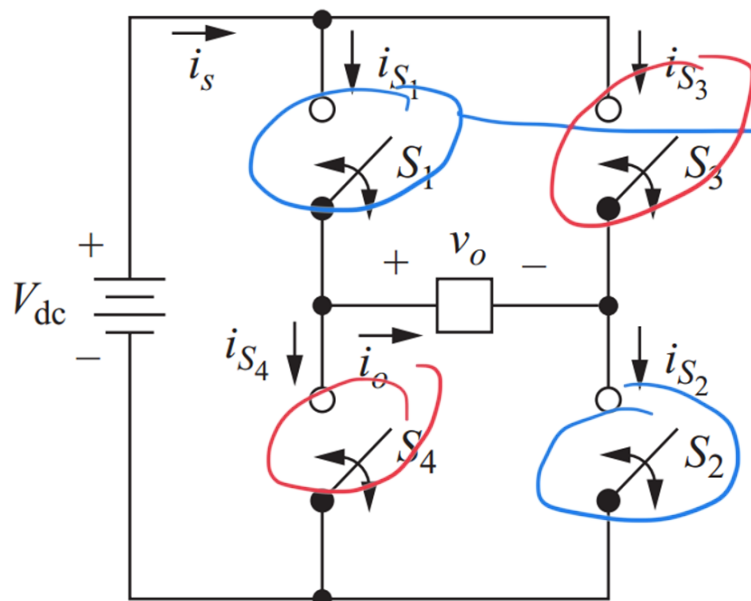
➤ **Switches should be bidirectional to allow both +/- currents**

Square-Wave Inverter

❑ **Objective:** Producing a square-wave output voltage from a DC source

❖ **RL Load:**

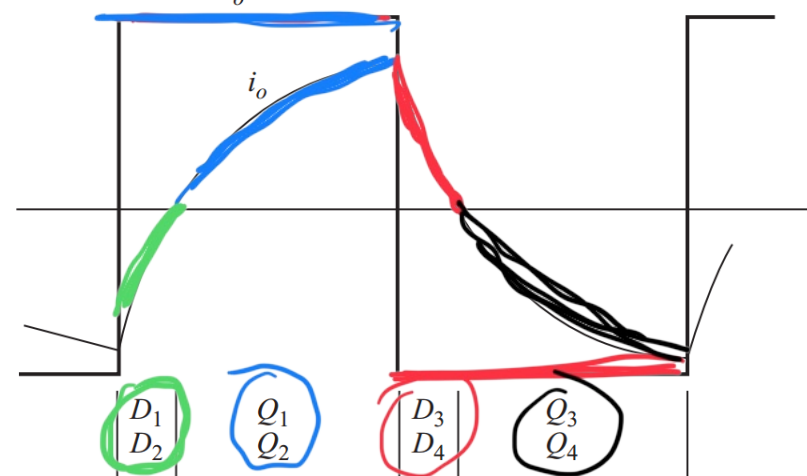
✓ **Using IGBTs and antiparallel diodes to make bidirectional switches**



S1 & S2 ON: $v_o = V_{dc}$

S3 & S4 ON: $v_o = -V_{dc}$

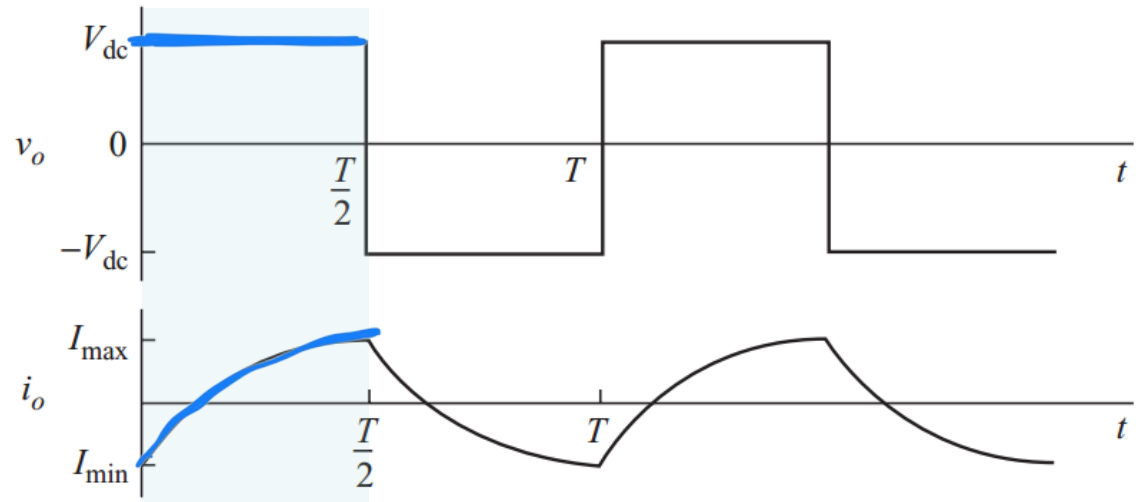
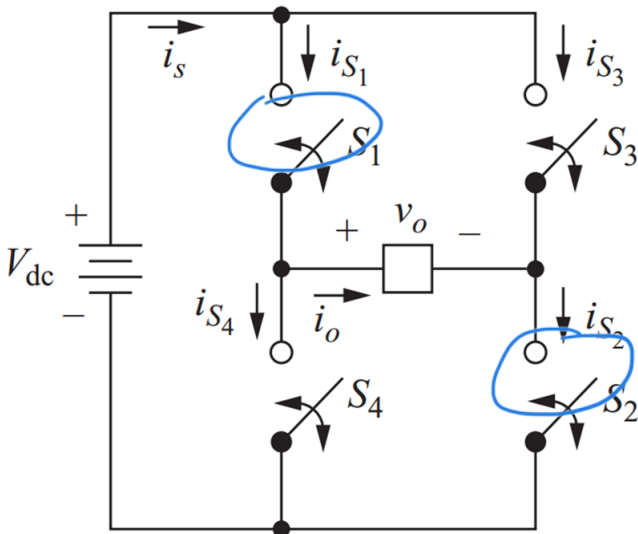
➤ **Switches should be bidirectional to allow both +/- currents**



Square-Wave Inverter

❏ **Objective:** Producing a square-wave output voltage from a DC source

❖ **RL Load:**



➤ **Current when** $v_o = V_{dc}$ **during** $0 < t < T/2$:

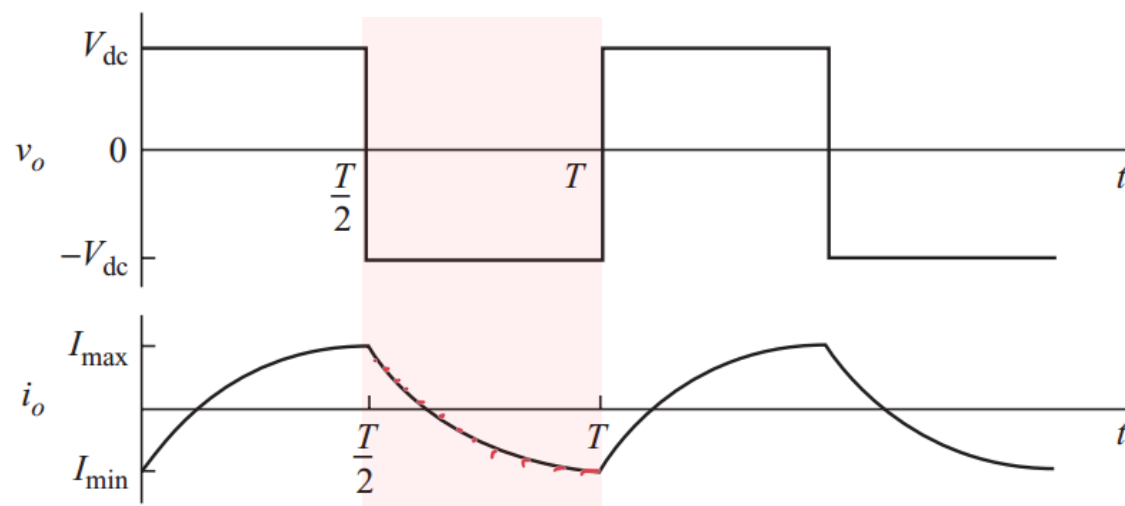
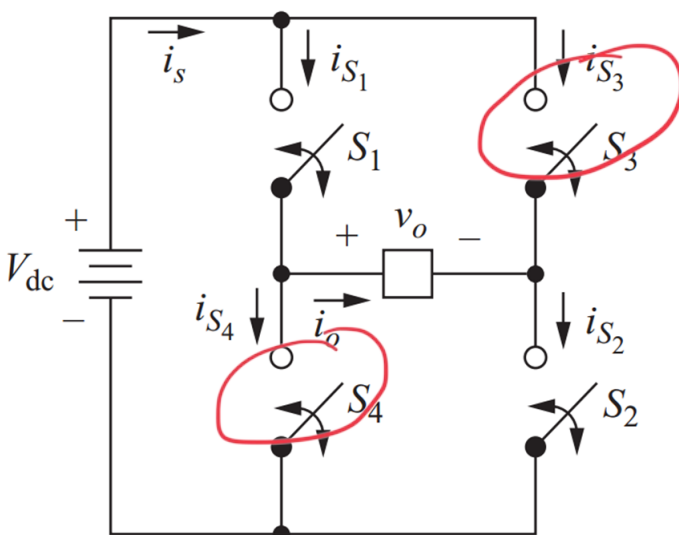
✓ **KVL:** $\underline{v_o = Ri_o(t) + L \frac{di_o(t)}{dt} = \sqrt{V_{dc}}}$



Square-Wave Inverter

❏ **Objective:** Producing a square-wave output voltage from a DC source

❖ **RL Load:**



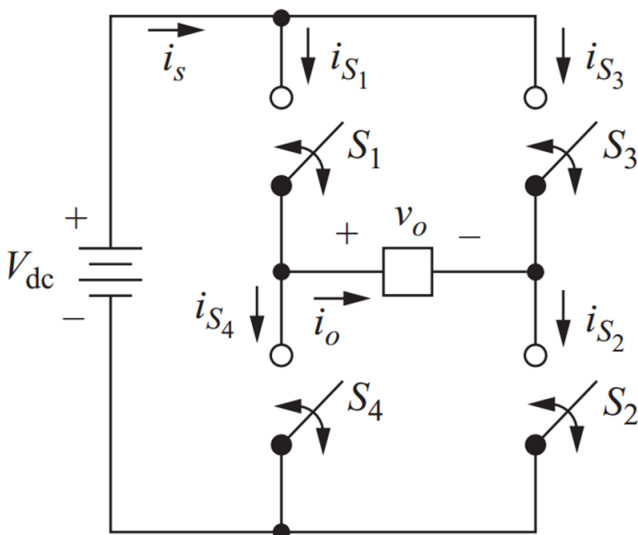
➤ **Current when $v_o = -V_{dc}$ during $T/2 < t < T$:**

✓ **KVL:** $v_o = Ri_o(t) + L \frac{di_o(t)}{dt} = -V_{dc}$

Square-Wave Inverter

❑ **Objective:** Producing a square-wave output voltage from a DC source

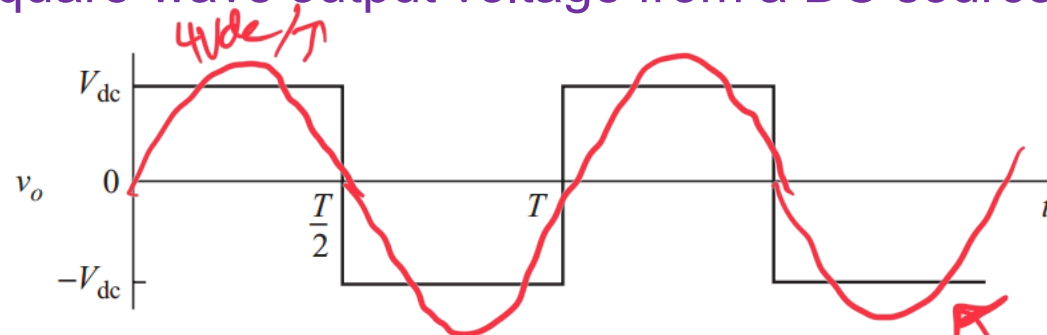
❖ **RL Load:**



Recall:

$$v_o(t) = \sum_{n=1}^{\infty} V_n \sin(n\omega_0 t + \theta_{v,n})$$

$$i_o(t) = \sum_{n=1}^{\infty} I_n \sin(n\omega_0 t + \theta_{i,n})$$



- In general, with an arbitrary (and possibly more complex) load, the Fourier series of the output voltage and current can be used to analyze the power, etc.
- The dc component (average value) is also zero!
- ❖ For a square wave, the Fourier series contains only the odd harmonics:

$$v_o(t) = \sum_{n \text{ odd}} \left(\frac{4V_{dc}}{n\pi} \right) \sin(n\omega_0 t)$$

$$a_0 = 0$$

$$a_n = 0$$

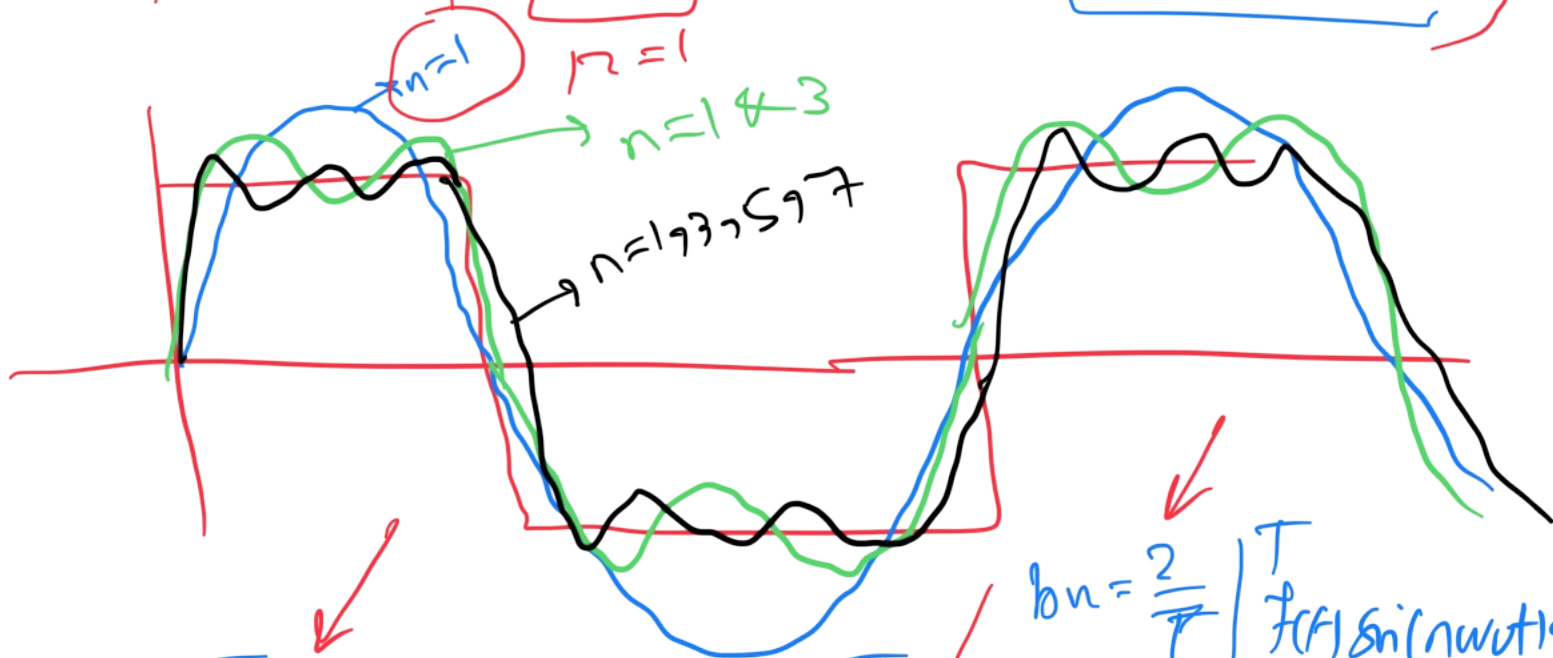
$$b_n = \frac{4V_{dc}}{n\pi}$$

$$\frac{4V_{dc}}{7\pi}$$

$f(t)$

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right]$$

$c_n \cos(n\omega_0 t + \theta_n)$



$$a_0 = \frac{1}{T} \int_0^T f(t) dt$$

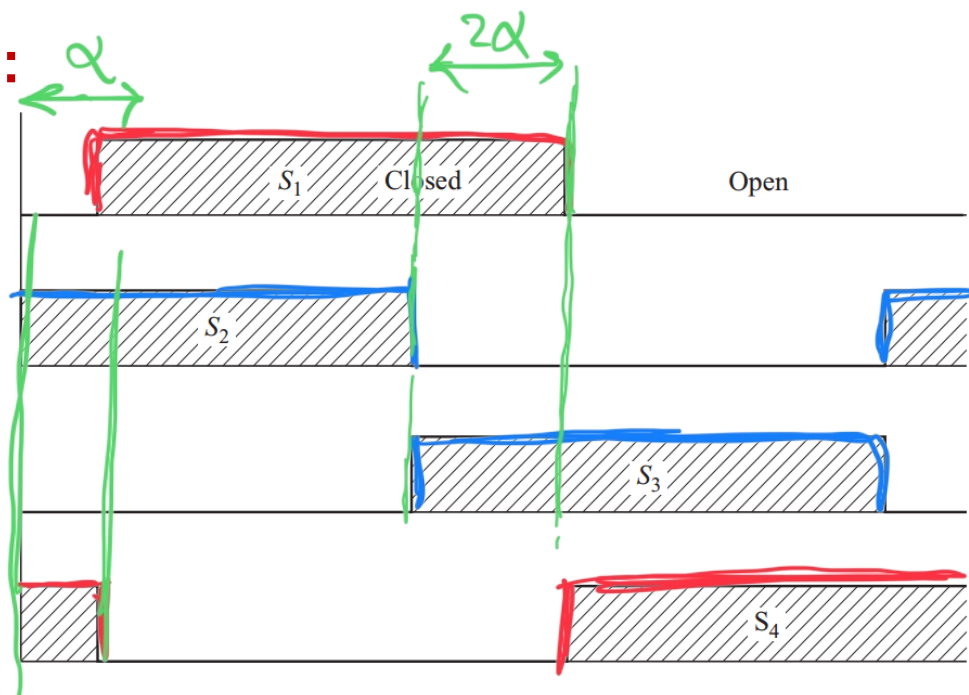
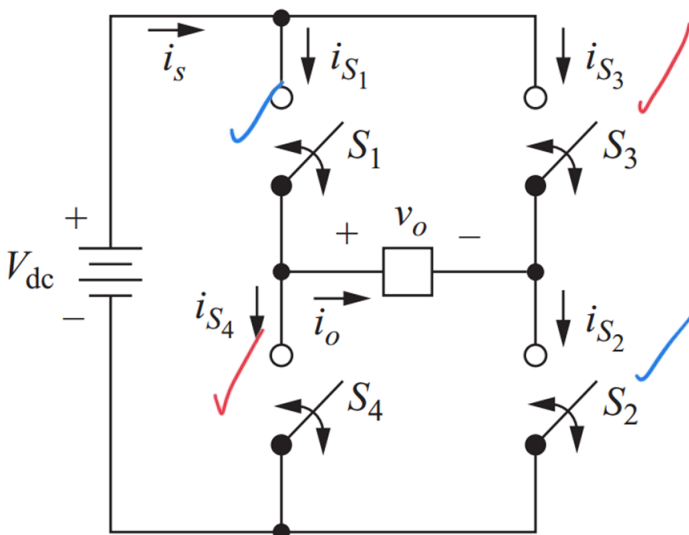
$$a_n = \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t) dt$$

Amplitude and Harmonic Control

❑ **Objective:** Controlling amplitude of harmonics using modified switching

❖ **Adding zero output voltage:**



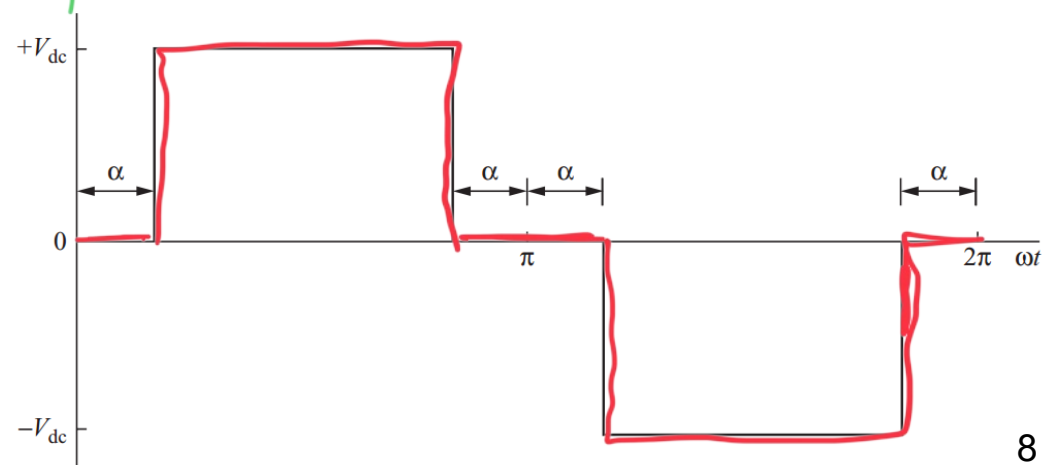
Switching Options

S1 & S2 ON: $v_o = V_{dc}$

S3 & S4 ON: $v_o = -V_{dc}$

S1 & S3 ON: $v_o = 0$

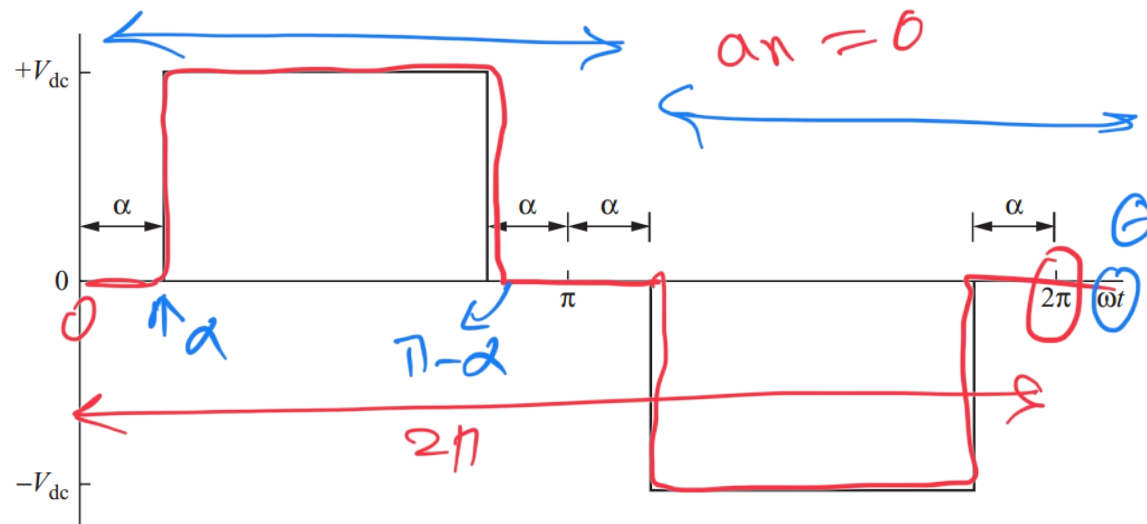
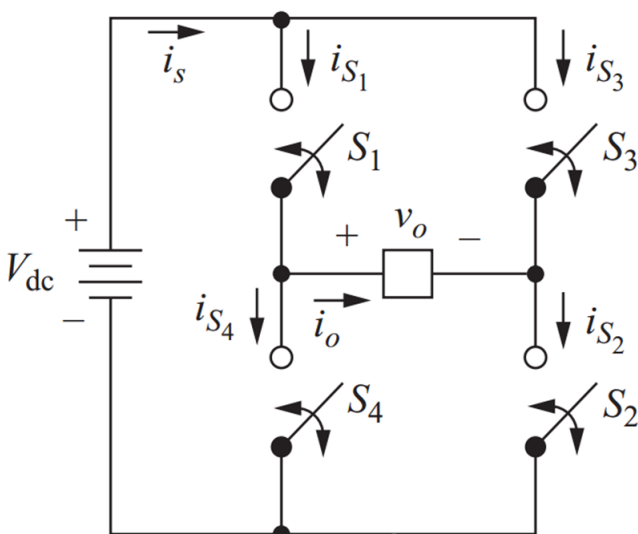
S2 & S4 ON: $v_o = 0$



Amplitude and Harmonic Control

❑ **Objective:** Controlling amplitude of harmonics using modified switching

❖ **Adding zero output voltage:**



$$v_o(t) = \sum_{n \text{ odd}} V_n \sin(n\omega_0 t)$$

$$V_n = \frac{2}{2\pi} \int_0^{2\pi} v_o(\omega_0 t) \sin(n\omega_0 t) d(\omega_0 t)$$

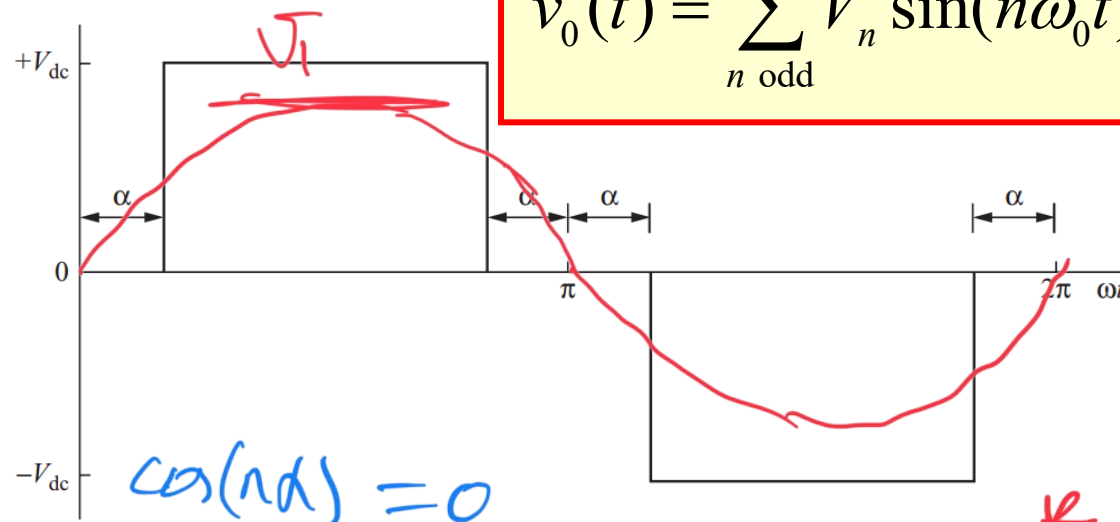
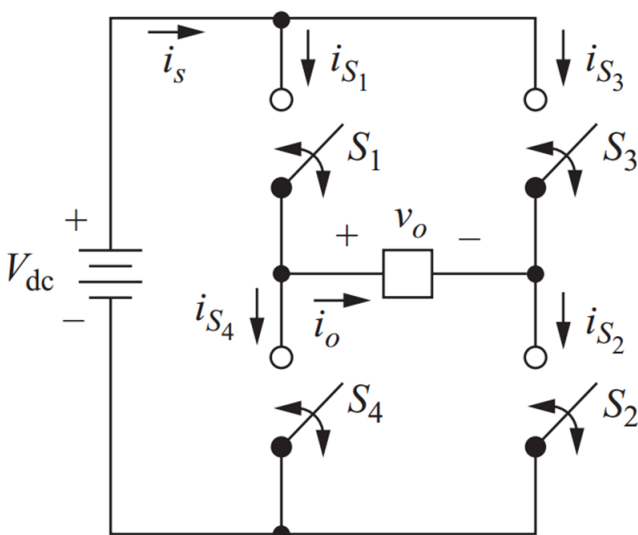
$$\Rightarrow V_n = \frac{2 \times 2}{2\pi} \int_{\alpha}^{\pi-\alpha} (V_{dc}) \sin(n\theta) d\theta = \frac{2}{\pi} \times \frac{1}{n} \omega_0(n\theta) \Big|_{\alpha}^{\pi-\alpha}$$

$$V_n = \left(\frac{4V_{dc}}{n\pi} \right) \cos(n\alpha)$$

Amplitude and Harmonic Control

❑ **Objective:** Controlling amplitude of harmonics using modified switching

❖ **Adding zero output voltage:**



✓ **Amplitude of fundamental frequency component is controllable using α**

$$V_1 = \frac{4V_{dc}}{\pi} \times \cos\alpha$$

$$V_n = \left(\frac{4V_{dc}}{n\pi} \right) \cos(n\alpha)$$

✓ **Amplitude of harmonics can also be controlled**



✓ **n -th harmonic can be eliminated if α is set:**

$$\alpha = \frac{90^\circ}{n}$$

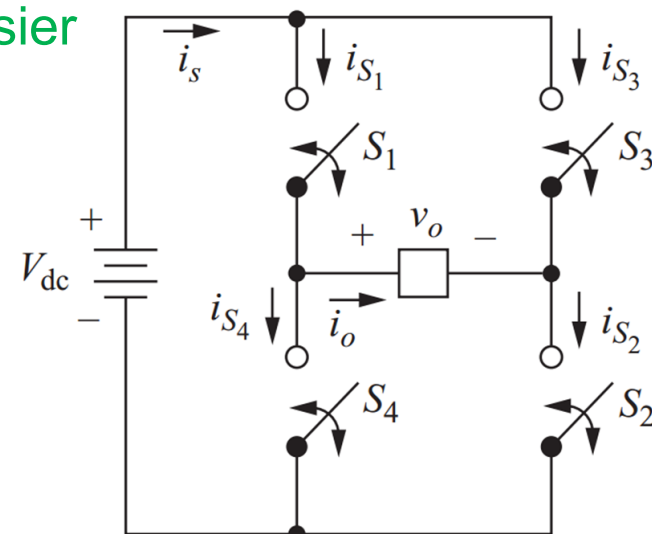
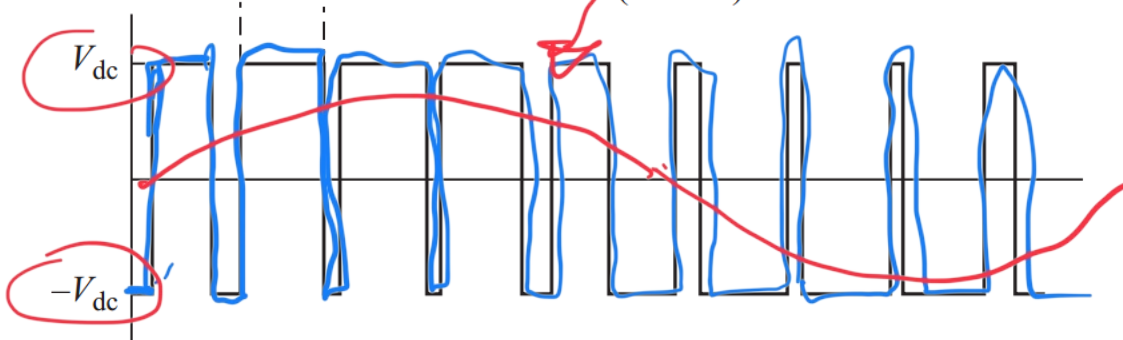
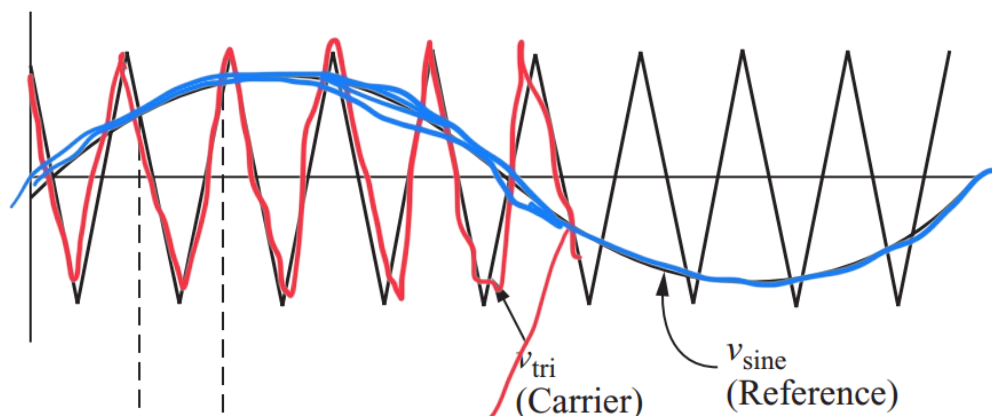
$$\frac{90^\circ}{3} = \alpha = 30^\circ \Rightarrow V_3 = 0$$

Pulse Width Modulation (PWM)

❑ **Objective:** PWM provides a way to decrease the THD of load current

✓ Compared to the square wave switching, the PWM harmonics will be at much higher frequencies, making filtering easier

❖ Bipolar Switching:

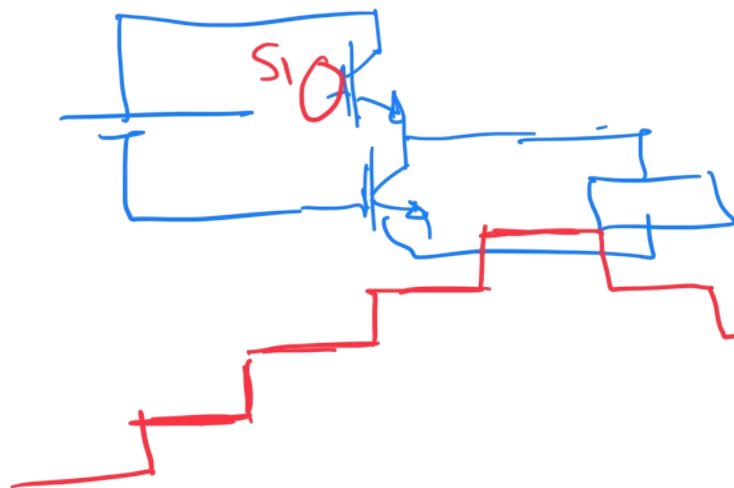
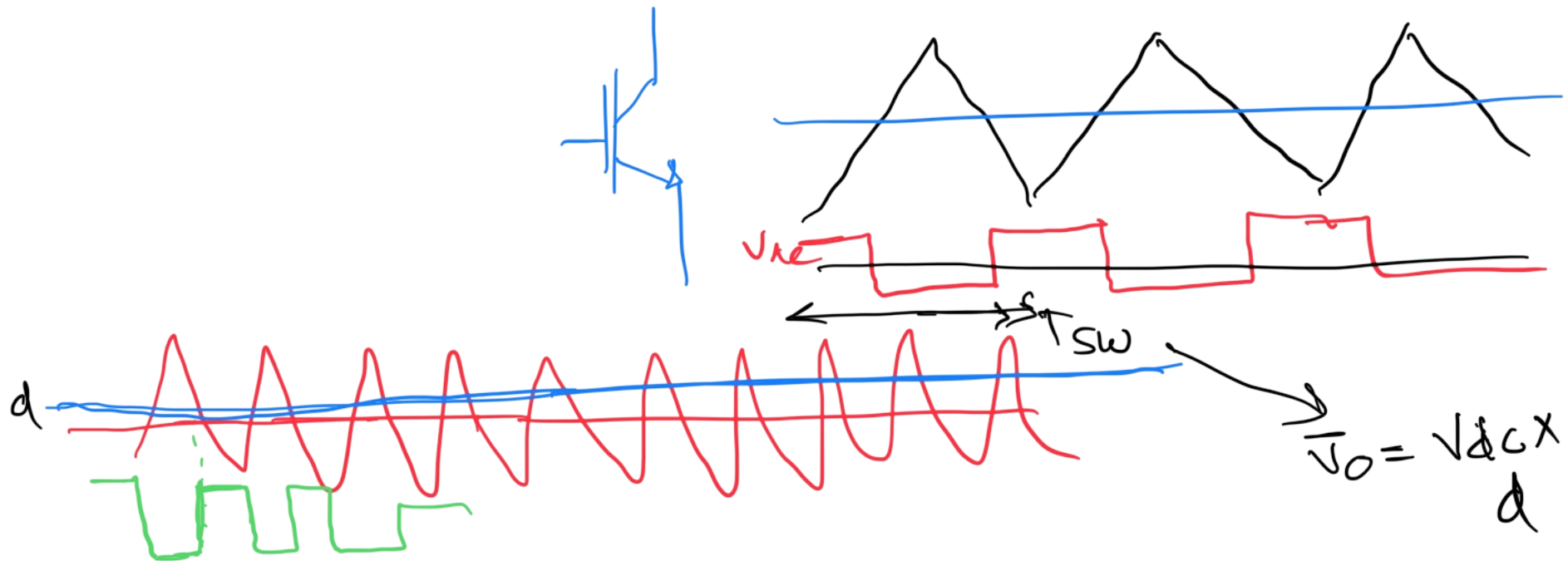


$v_{\sin} > v_{tri}$ **S1 & S2 ON**

$v_o = +V_{dc}$

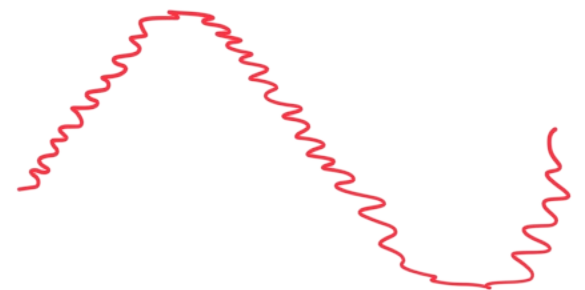
$v_{\sin} < v_{tri}$ **S3 & S4 ON**

$v_o = -V_{dc}$



$$S_1 = 1 \rightarrow V_O = V_{dc}$$

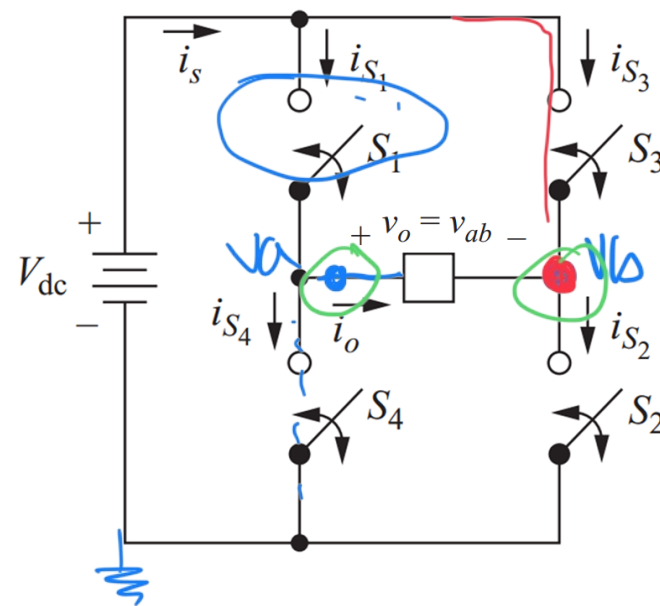
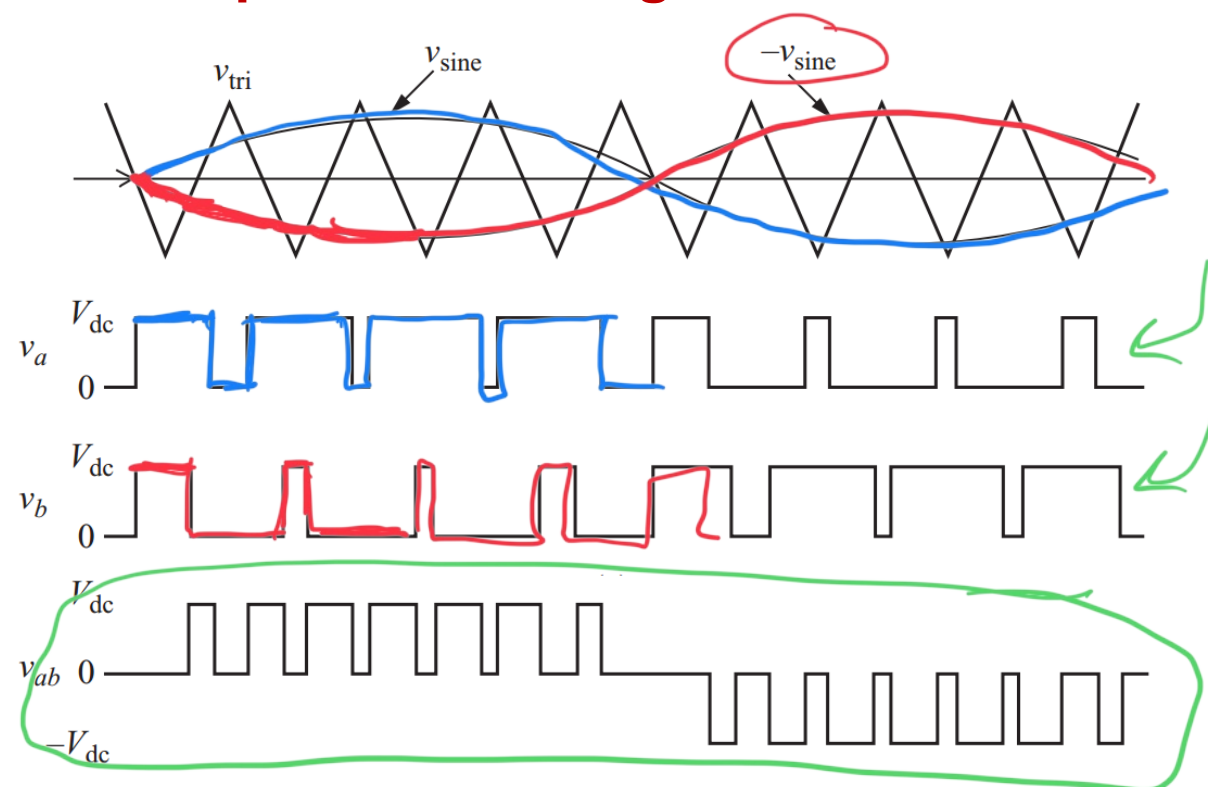
$$S_1 = 0 \rightarrow V_O = 0$$



Pulse Width Modulation (PWM)

❑ **Objective:** PWM provides a way to decrease the THD of load current

❖ Unipolar Switching:



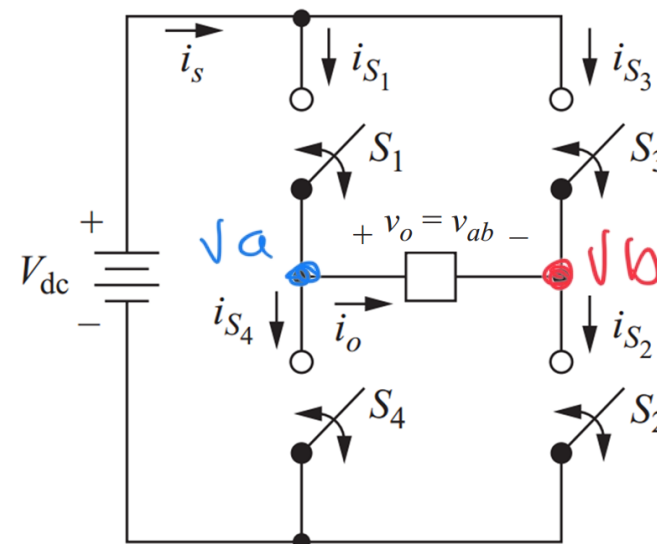
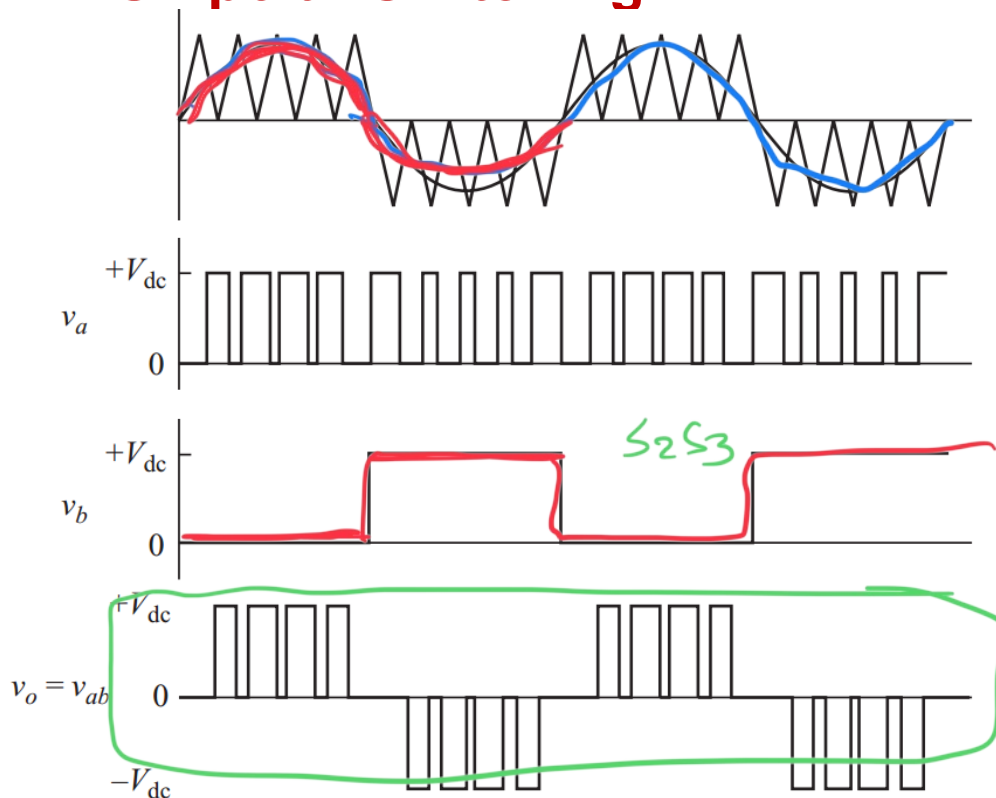
✓ The switch pair (**S1,S4**) are complementary; similarly pair (**S2,S3**)

$$\left\{ \begin{array}{l} v_{sin} > v_{tri} \\ v_{sin} < v_{tri} \end{array} \right. \begin{array}{l} \text{S1 ON} \\ \text{S4 ON} \end{array} \left\{ \begin{array}{l} v_a = V_{dc} \\ v_a = 0 \end{array} \right. \quad \left\{ \begin{array}{l} -v_{sin} > v_{tri} \\ -v_{sin} < v_{tri} \end{array} \right. \begin{array}{l} \text{S3 ON} \\ \text{S2 ON} \end{array} \left\{ \begin{array}{l} v_b = V_{dc} \\ v_b = 0 \end{array} \right.$$

Pulse Width Modulation (PWM)

❑ **Objective:** PWM provides a way to decrease the THD of load current

❖ Unipolar Switching:



✓ The switch pair (**S1, S4**) are complementary at carrier frequency; similarly pair (**S2, S3**) at reference frequency

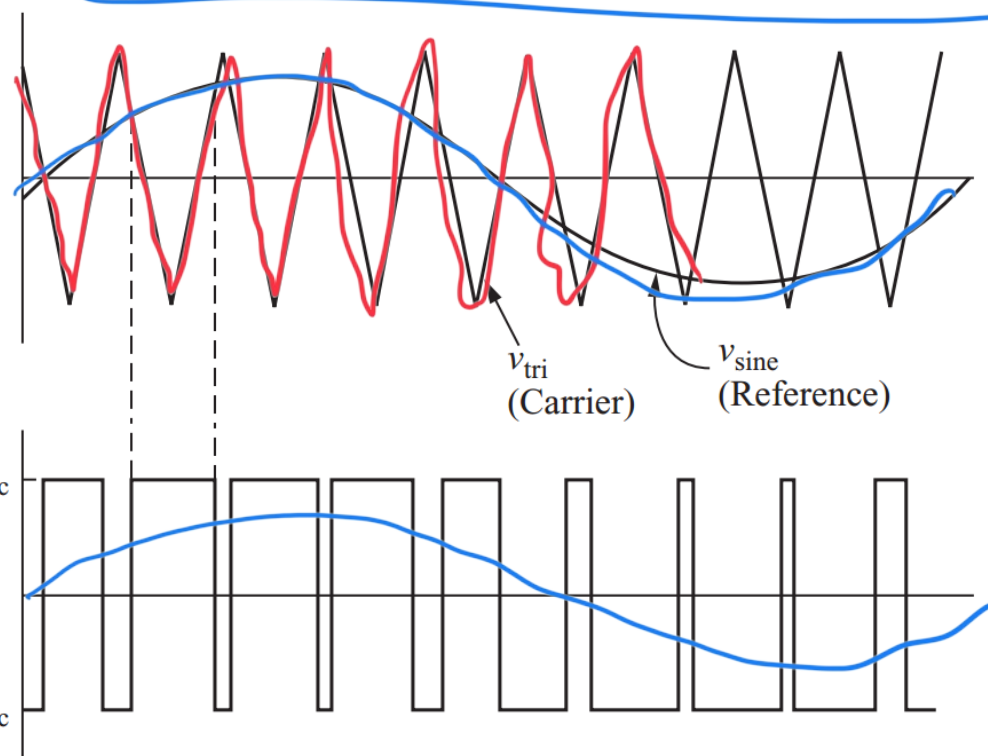
$$\left\{ \begin{array}{l} v_{\sin} > v_{\text{tri}} \\ v_{\sin} < v_{\text{tri}} \end{array} \right. \quad \begin{array}{l} \text{S1 ON } v_a = V_{dc} \\ \text{S4 ON } v_a = 0 \end{array}$$

$$\left\{ \begin{array}{l} v_{\sin} < 0 \\ v_{\sin} > 0 \end{array} \right. \quad \begin{array}{l} \text{S3 ON } v_b = V_{dc} \\ \text{S2 ON } v_b = 0 \end{array}$$

Pulse Width Modulation (PWM)

❑ **Objective:** PWM provides a way to decrease the THD of load current

❖ **Frequency Modulation Ratio (m_f):**



$$m_f = \frac{f_{\text{carrier}}}{f_{\text{reference}}} = \frac{f_{\text{tri}}}{f_{\text{sine}}}$$

- ✓ Increasing the carrier frequency (increasing m_f) increases the frequencies at which the harmonics occur
- ✓ With the harmonics at higher frequency, simpler low-pass filters can be used to remove them; however, the inverter switches will also have higher losses

➤ The Fourier series of the PWMed output voltage has a fundamental frequency which is the same as the reference signal

Pulse Width Modulation (PWM)

❑ **Objective:** PWM provides a way to decrease the THD of load current

❖ **Amplitude Modulation Ratio (m_a):**

$$m_a = \frac{V_{m,\text{reference}}}{V_{m,\text{carrier}}} = \frac{V_{m,\text{sine}}}{V_{m,\text{tri}}}$$

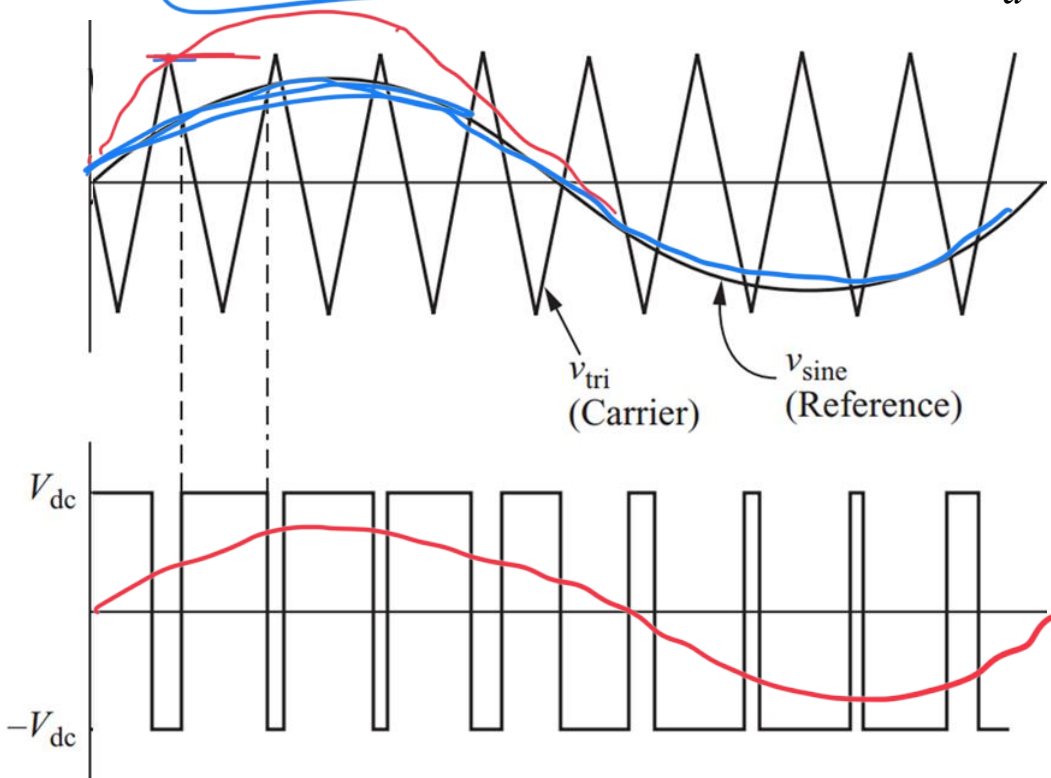
✓ As long as $m_a \leq 1$:

$$V_o^1 = m_a V_{dc}$$

✓ Amplitude of the fundamental frequency component of output voltage is controlled and changed linearly with m_a

➤ The Fourier series of the PWM output voltage has a fundamental frequency which is the same as the reference signal

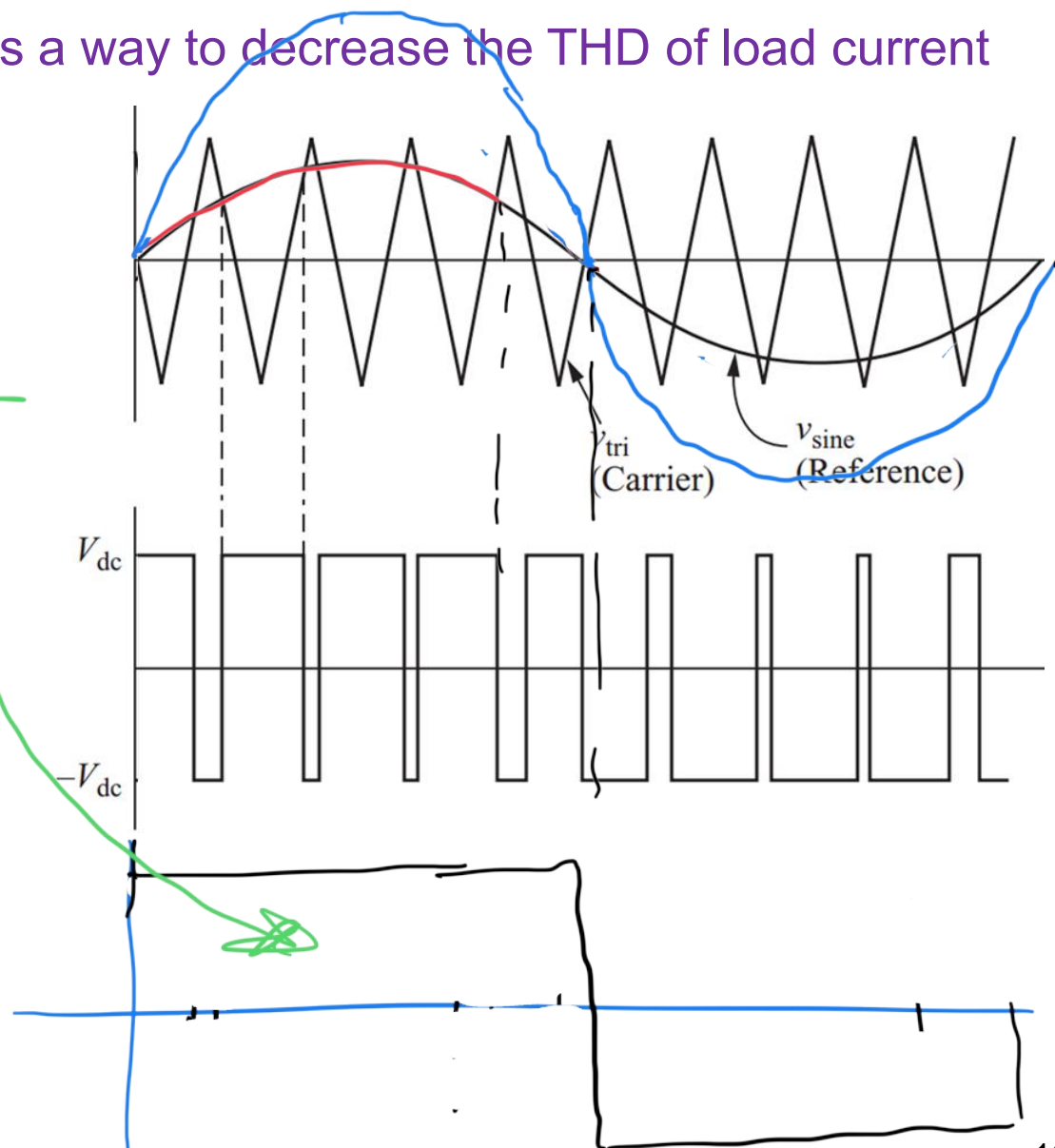
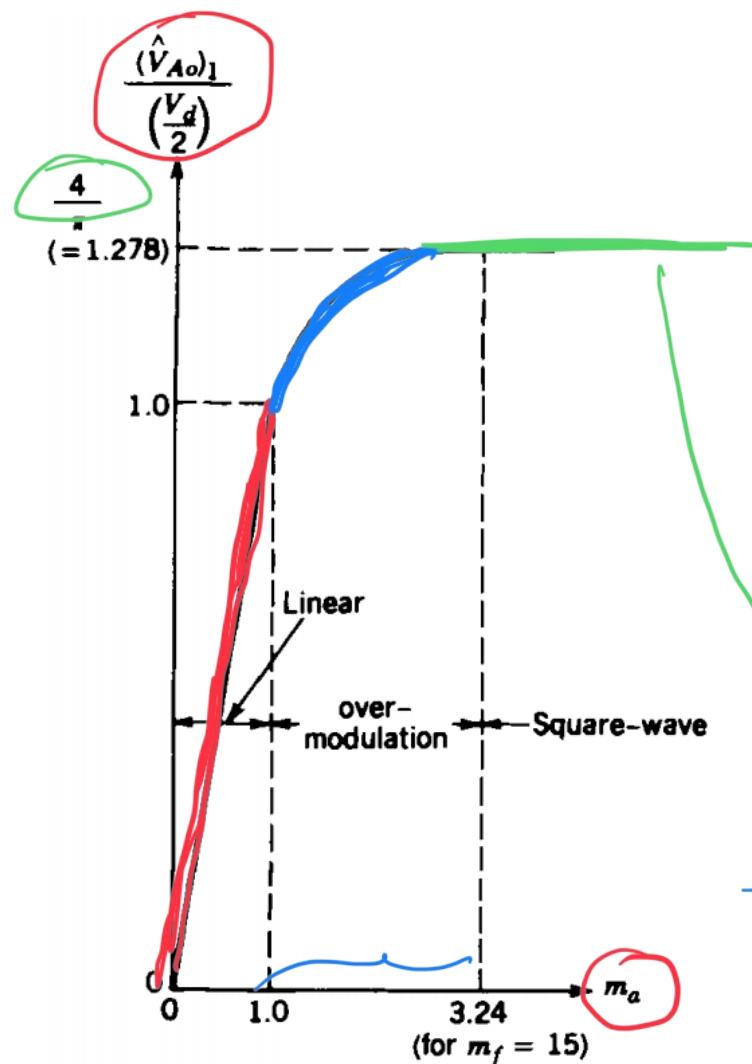
➤ With over modulation $m_a > 1$ relation becomes nonlinear



Three-Phase PWM Inverters

❑ **Objective:** PWM provides a way to decrease the THD of load current

❖ **Overmodulation:**

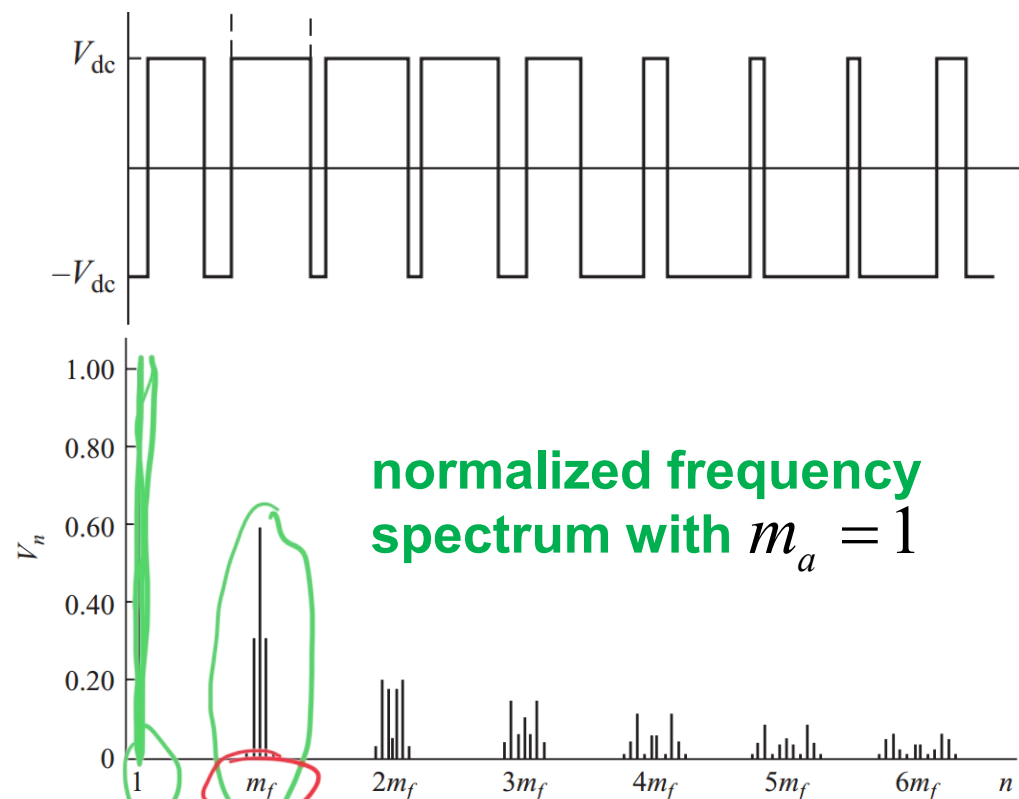


Pulse Width Modulation (PWM)

❑ **Objective:** PWM provides a way to decrease the THD of load current

❖ Harmonic Content with Bipolar Switching:

- ✓ m_f is chosen to be an odd integer
- ✓ The harmonic content are not a function of m_f if it is sufficiently large; but they are a function of m_a
- ✓ The harmonics start at around m_f



Normalized Fourier Coefficients V_n/V_{dc} for Bipolar PWM

	$m_a=1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n=1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n=m_f$	0.60	0.71	0.82	0.92	1.01	1.08	1.15	1.20	1.24	1.27
$n=m_f \pm 2$	0.32	0.27	0.22	0.17	0.13	0.09	0.06	0.03	0.02	0.00

Pulse Width Modulation (PWM)

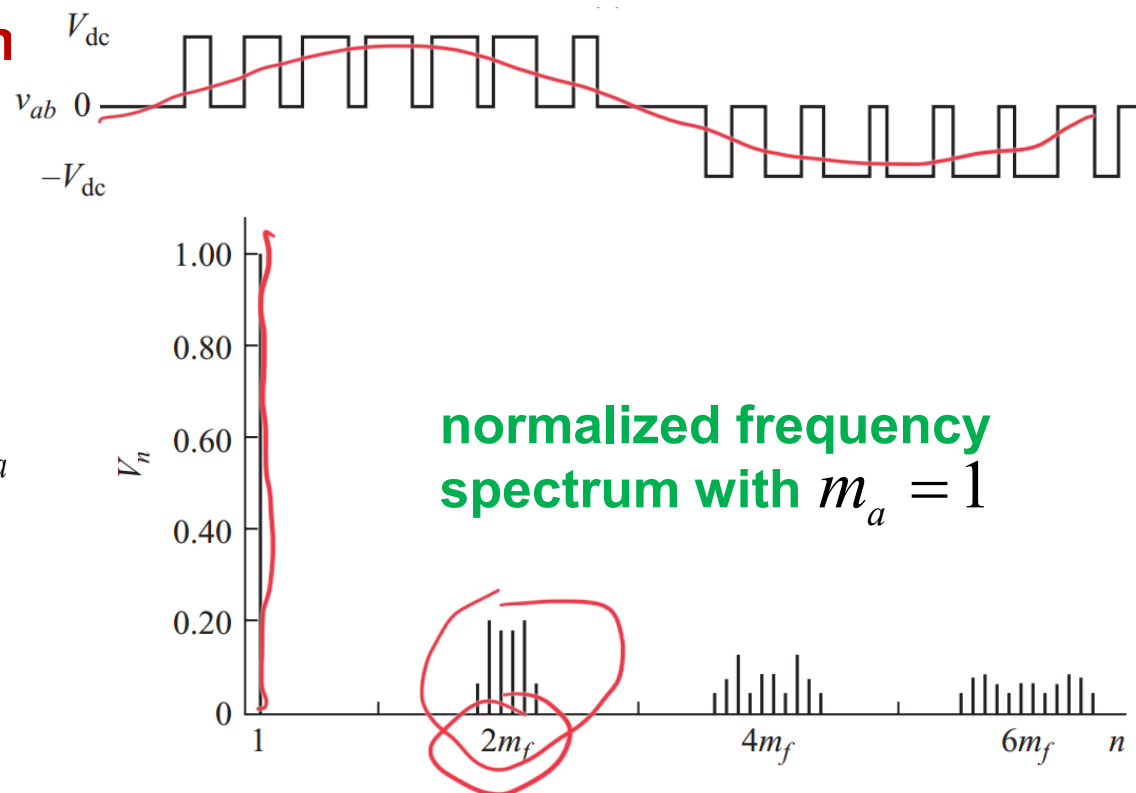
❑ **Objective:** PWM provides a way to decrease the THD of load current

❖ Harmonic Content with Unipolar Switching:

✓ m_f is chosen to be an even integer

✓ The are dependent on m_a

✓ The harmonics start at around $2m_f$



Normalized Fourier Coefficients V_n/V_{dc} for Unipolar PWM

	$m_a=1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n=1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n=2m_f \pm 1$	0.18	0.25	0.31	0.35	0.37	0.36	0.33	0.27	0.19	0.10
$n=2m_f \pm 3$	0.21	0.18	0.14	0.10	0.07	0.04	0.02	0.01	0.00	0.00

Three-Phase Inverters

❑ **Objective:** Producing 3-phase AC outputs from a DC source

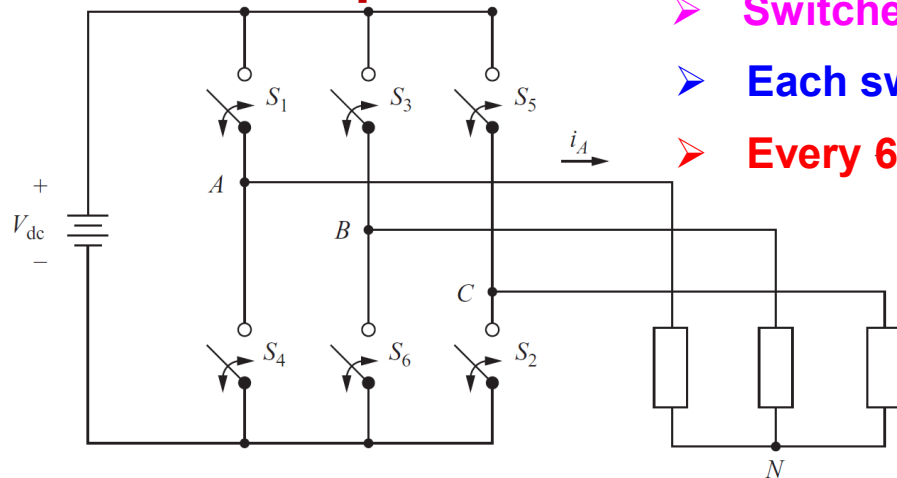
❖ Six-Step Inverter:

➤ Switches in each leg are complementary

➤ Each switch has a duty cycle of 50%

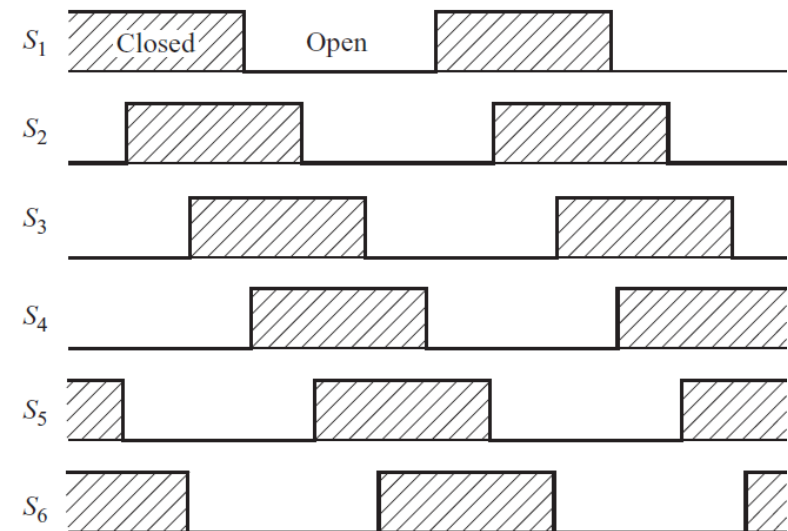
➤ Every 60° one switching occurs

$$\begin{cases} S_4 = \bar{S}_1 \\ S_6 = \bar{S}_3 \\ S_2 = \bar{S}_5 \end{cases}$$



$$\begin{cases} v_{AG} = v_{AN} + v_{NG} \\ v_{BG} = v_{BN} + v_{NG} \\ v_{CG} = v_{CN} + v_{NG} \end{cases} \Rightarrow v_{NG} = \frac{1}{3}(v_{AG} + v_{BG} + v_{CG})$$

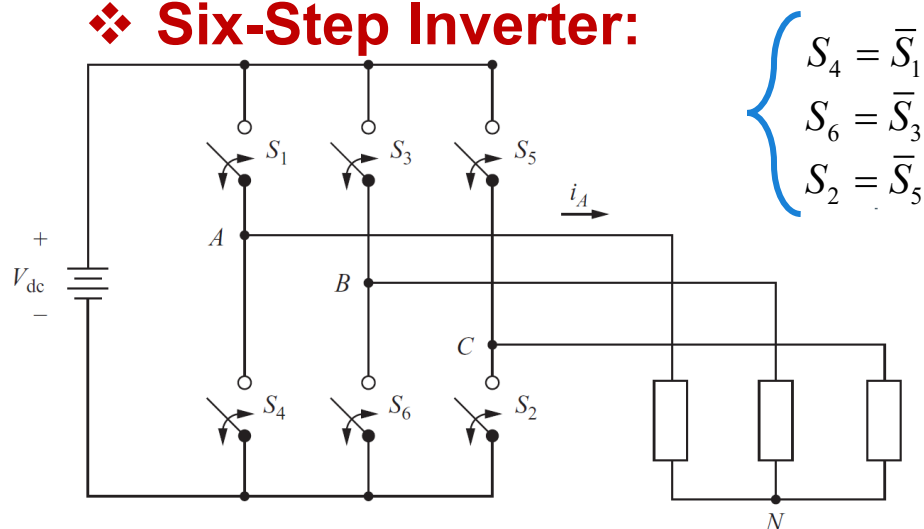
$$\begin{cases} v_{AG} = S_1 \cdot V_{dc} \\ v_{BG} = S_3 \cdot V_{dc} \\ v_{CG} = S_5 \cdot V_{dc} \end{cases} \Rightarrow \begin{bmatrix} v_{AN} \\ v_{BN} \\ v_{CN} \end{bmatrix} = V_{dc} \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} \begin{bmatrix} S_1 \\ S_3 \\ S_5 \end{bmatrix}$$



Three-Phase Inverters

❏ **Objective:** Producing 3-phase AC outputs from a DC source

❖ Six-Step Inverter:

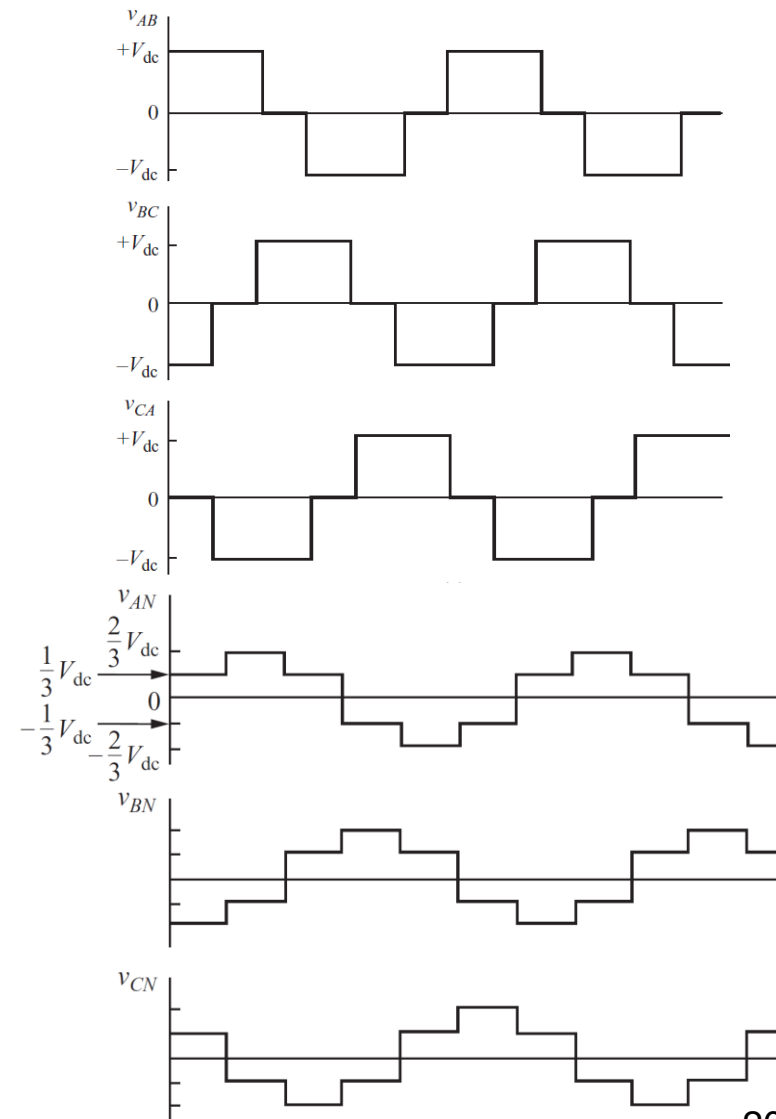


✓ Output voltage has a fundamental frequency equal to the switching frequency

✓ Harmonics are of the order $n = 6k \pm 1$
with $k \in \{1, 2, \dots\} \Rightarrow n \in \{1, 5, 7, 11, 13, \dots\}$

$$V_n^{LL} = \frac{4V_{dc}}{n\pi} \left| \cos\left(n \frac{\pi}{6}\right) \right|$$

$$V_n^{LN} = \frac{2V_{dc}}{3n\pi} \left| 2 + \cos\left(n \frac{\pi}{3}\right) - \cos\left(n \frac{2\pi}{3}\right) \right|$$



Three-Phase Inverters

❑ **Objective:** Producing 3-phase AC outputs from a DC source

❖ **Six-Step Inverter:**

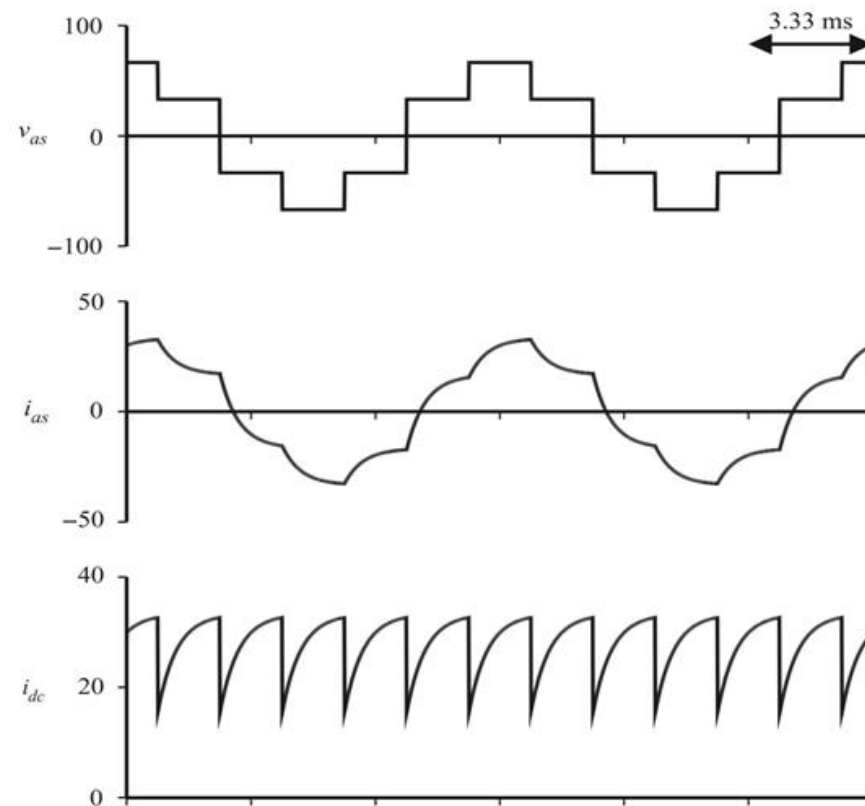
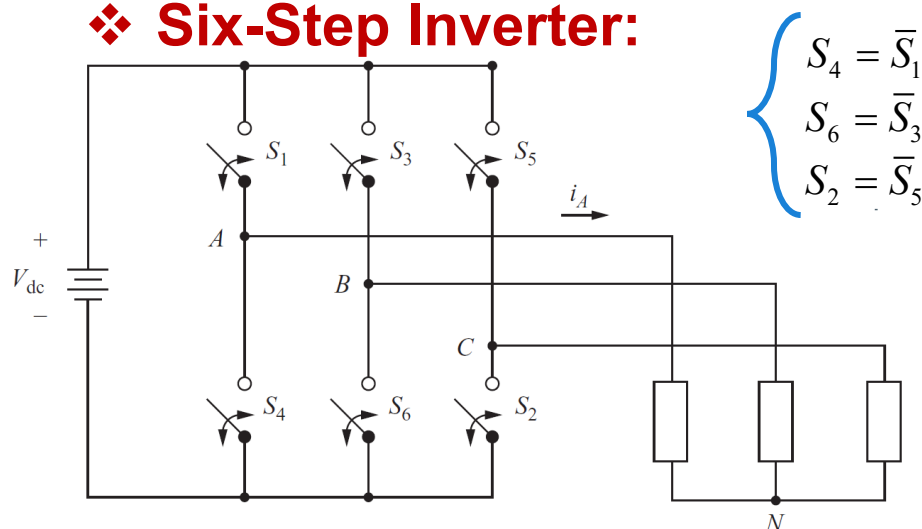
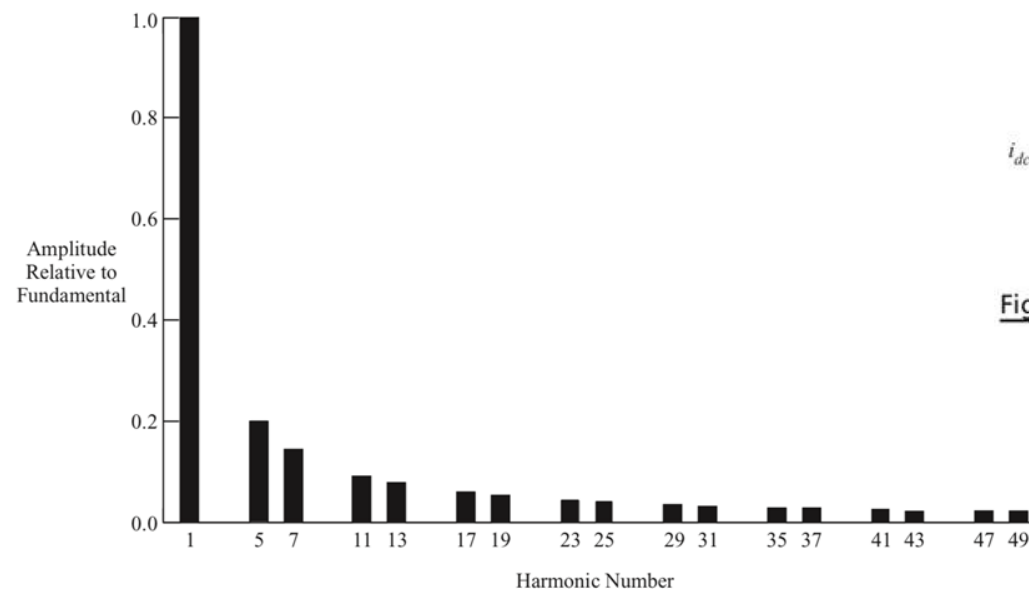


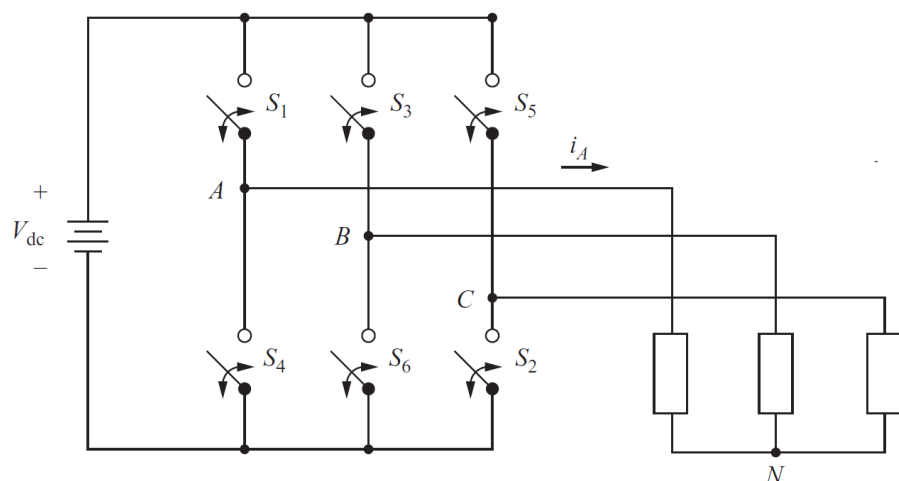
Figure. Voltage and current waveforms for a six-stepped converter feeding an RL load.



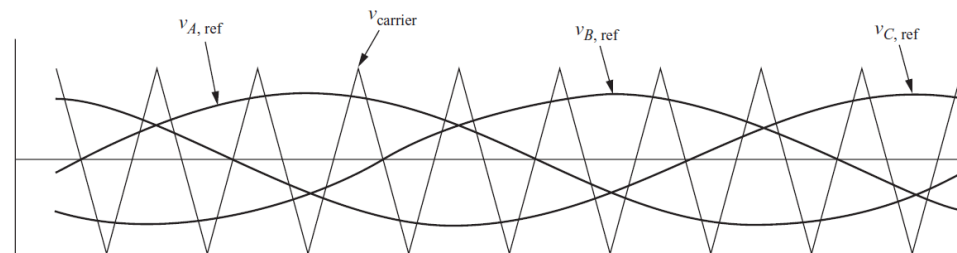
Three-Phase PWM Inverters

❑ **Objective:** Producing 3-phase AC outputs from a DC source with PWM

❖ 3-Phase PWM Inverter:



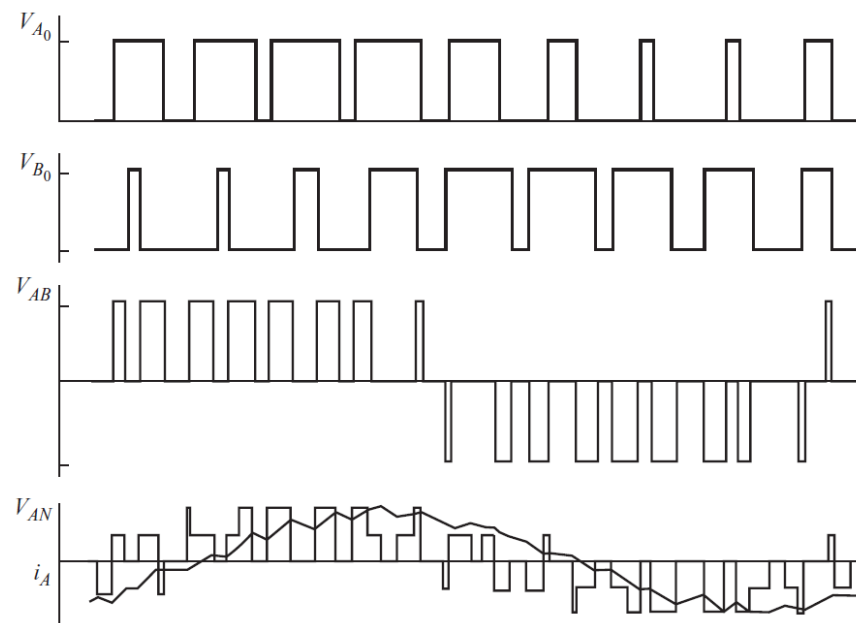
➤ Similar to the single-phase PWM inverter, a sinusoidal reference wave (for each phase) is compared with a triangular carrier wave



➤ Switches in each leg are complementary

➡ $S_4 = \bar{S}_1$, $S_6 = \bar{S}_3$, $S_2 = \bar{S}_5$

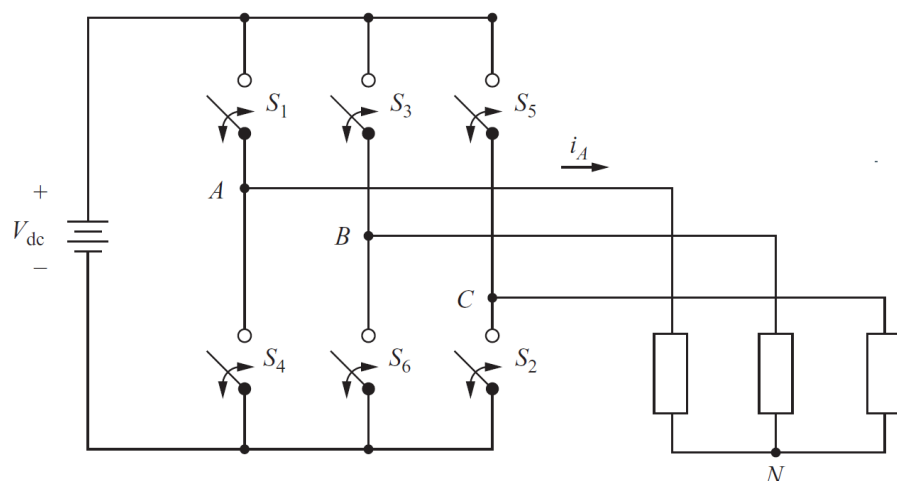
$$\left\{ \begin{array}{ll} v_a > v_{tri} & \text{S1 ON} \\ v_b > v_{tri} & \text{S3 ON} \\ v_c > v_{tri} & \text{S5 ON} \end{array} \right. \left\{ \begin{array}{ll} v_a < v_{tri} & \text{S4 ON} \\ v_b < v_{tri} & \text{S6 ON} \\ v_c < v_{tri} & \text{S2 ON} \end{array} \right.$$



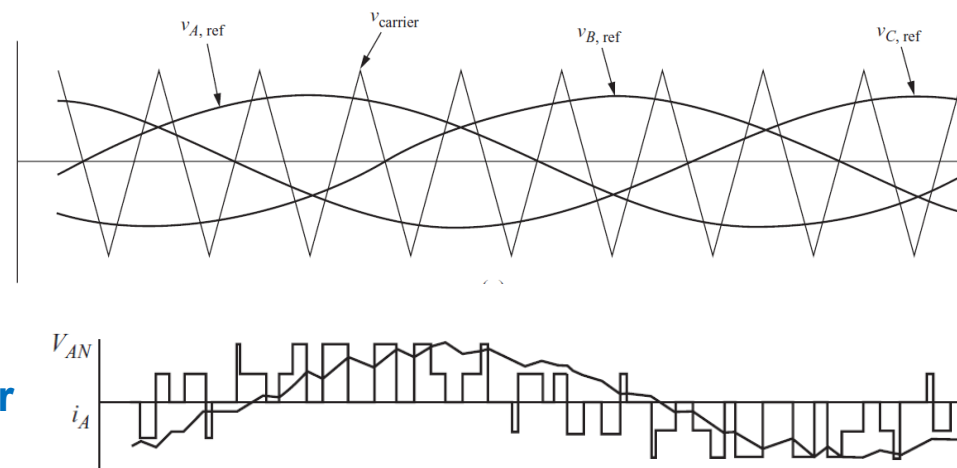
Three-Phase PWM Inverters

❑ **Objective:** Producing 3-phase AC outputs from a DC source with PWM

❖ 3-Phase PWM Inverter:



➤ Similar to the single-phase PWM inverter, a sinusoidal reference wave (for each phase) is compared with a triangular carrier wave



➤ Harmonics will be minimized if the carrier frequency is chosen to be an **odd triple** multiple of the reference frequency, i.e.,

$$m_f \in \{3, 9, 15, \dots\}$$

Normalized Amplitudes V_n / V_{dc} for Line-to-Line Three-Phase PWM Voltages

	$m_a=1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n = 1$	0.866	0.779	0.693	0.606	0.520	0.433	0.346	0.260	0.173	0.087
$n = m_f \pm 2$	0.275	0.232	0.190	0.150	0.114	0.081	0.053	0.030	0.013	0.003
$n = 2m_f \pm 1$	0.157	0.221	0.272	0.307	0.321	0.313	0.282	0.232	0.165	0.086

Three-Phase PWM Inverters

❑ **Objective:** Producing 3-phase AC outputs from a DC source with PWM

❖ 3-Phase PWM Inverter:

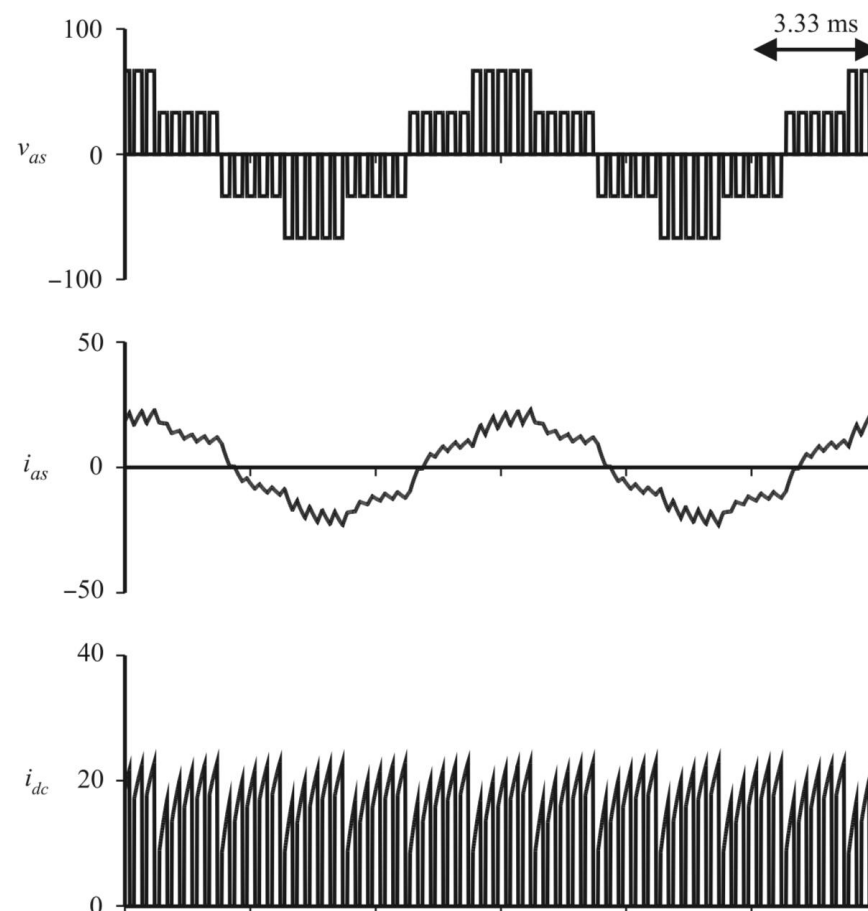
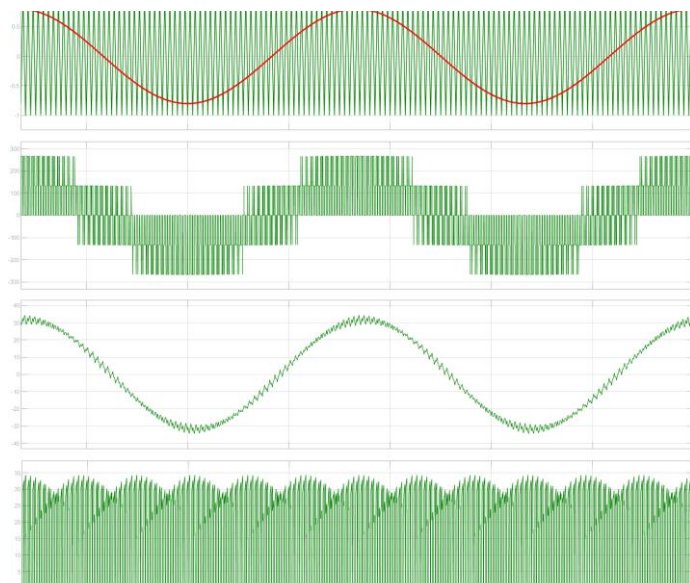
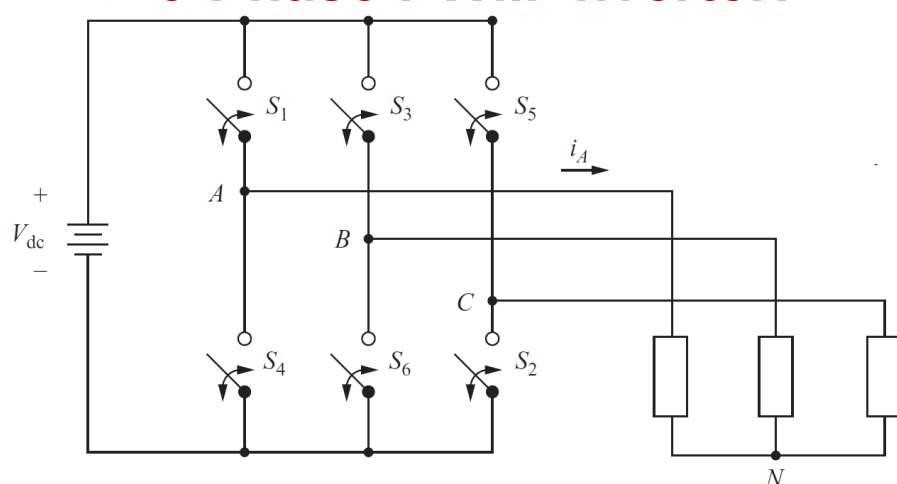


Figure 12.4-3. Voltage and current waveforms for six-step modulated converter feeding an RL load.

Average-Value Model (AVM)

➤ In the AVMs, the switching ripples in the ac and dc variables are ignored

$$\left\{ \begin{array}{l} v_{dc} = \bar{v}_{dc} + \tilde{v}_{dc} \\ i_{dc} = \bar{i}_{dc} + \tilde{i}_{dc} \end{array} \right. \Rightarrow \bar{v}_{dc} = \frac{1}{T_{sw}} \int_{t-T_{sw}}^t v_{dc} d\tau \quad \bar{i}_{dc} = \frac{1}{T_{sw}} \int_{t-T_{sw}}^t i_{dc} d\tau$$

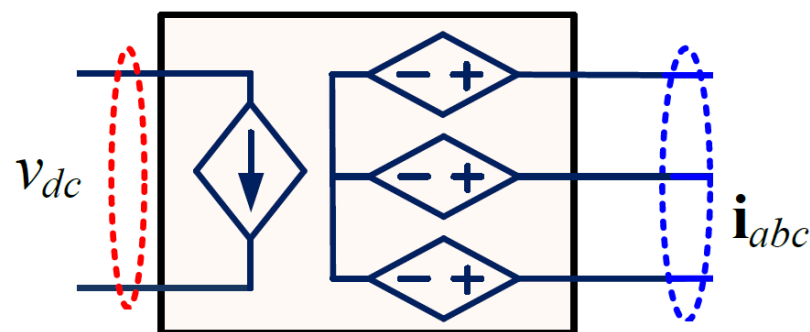
$$\left\{ \begin{array}{l} \mathbf{v}_{abc} = \mathbf{v}_{abc}^1 + \sum_n \mathbf{v}_{abc}^n \\ \mathbf{i}_{abc} = \mathbf{i}_{abc}^1 + \sum_n \mathbf{i}_{abc}^n \end{array} \right. \Rightarrow \mathbf{v}_{abc}^1 = \mathbf{m}_{abc} \times \frac{\bar{v}_{dc}}{2}$$

$$\mathbf{m}_{abc} = \begin{bmatrix} m_a \\ m_b \\ m_c \end{bmatrix} = M \times \begin{bmatrix} \cos(\theta_e) \\ \cos(\theta_e - 2\pi/3) \\ \cos(\theta_e + 2\pi/3) \end{bmatrix} \quad M = \frac{\|\mathbf{v}_{abc}^1\|}{v_{dc}/2}$$

$$\bar{i}_{dc} = \frac{1}{2} \times (m_a i_a + m_b i_b + m_c i_c)$$

Average-Value Model (AVM)

- In the AVMs, the switching ripples in the ac and dc variables are ignored
- In the AVMs, the switching devices are replaced with dependent voltage/current sources



$$i_{dc} = \frac{1}{2} \times (m_a i_a + m_b i_b + m_c i_c)$$

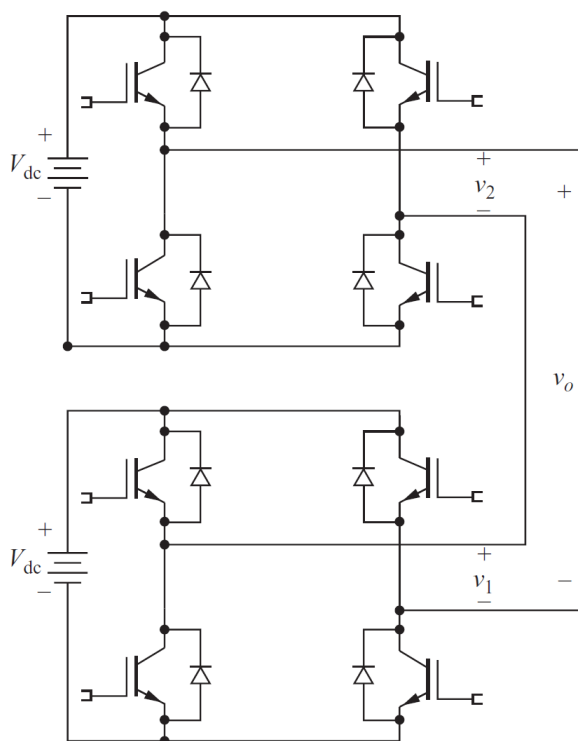
$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} m_a \\ m_b \\ m_c \end{bmatrix} \times \frac{v_{dc}}{2}$$

$$\begin{bmatrix} m_a \\ m_b \\ m_c \end{bmatrix} = M \times \begin{bmatrix} \cos(\theta_e) \\ \cos(\theta_e - 2\pi/3) \\ \cos(\theta_e + 2\pi/3) \end{bmatrix}$$

Multi-level Inverters

❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

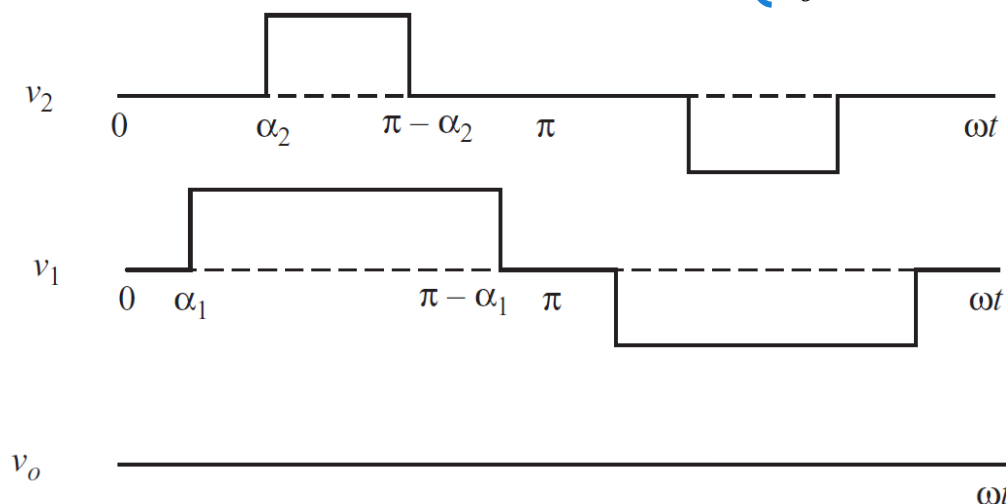
❖ **With 2 Independent Sources:**



$$v_k(t) = \sum_{n \text{ odd}} V_{n,k} \sin(n\omega_0 t)$$

$$V_{n,k} = \left(\frac{4V_{dc}}{n\pi} \right) \cos(n\alpha_k)$$

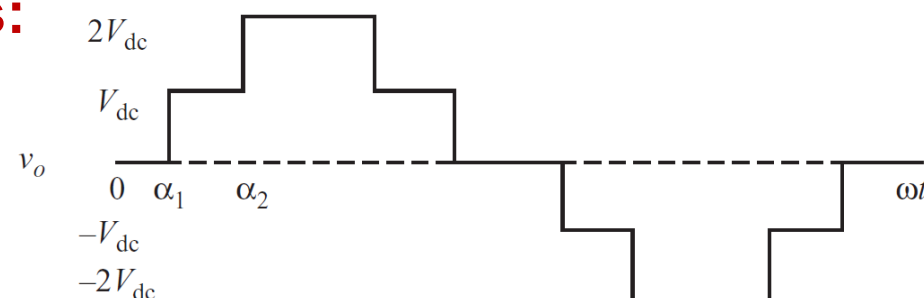
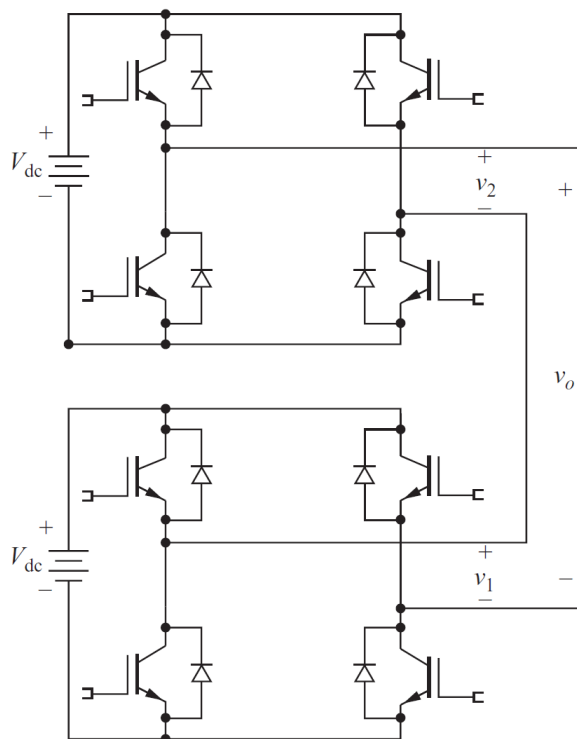
- Each H-bridge can output $+V_{dc}$, 0 or $-V_{dc}$
- The total output voltage can be any combination of these individual output levels
- Possible output levels for this two-source inverter



Multi-level Inverters

❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

❖ **With 2 Independent Sources:**



➤ **Fourier series of the total output voltage:**

$$v_o(t) = \sum_{n \text{ odd}} V_n \sin(n\omega_0 t)$$



$$V_n = \left(\frac{4V_{dc}}{n\pi} \right) [\cos(n\alpha_1) + \cos(n\alpha_2)]$$

➤ **How to eliminate the m th harmonic?**

➤ **Is there any other condition to meet?**

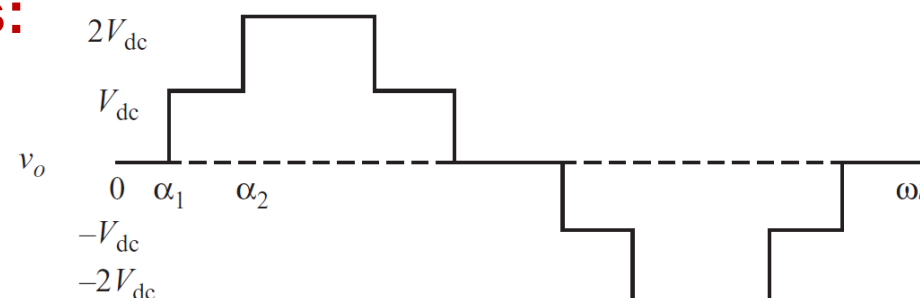
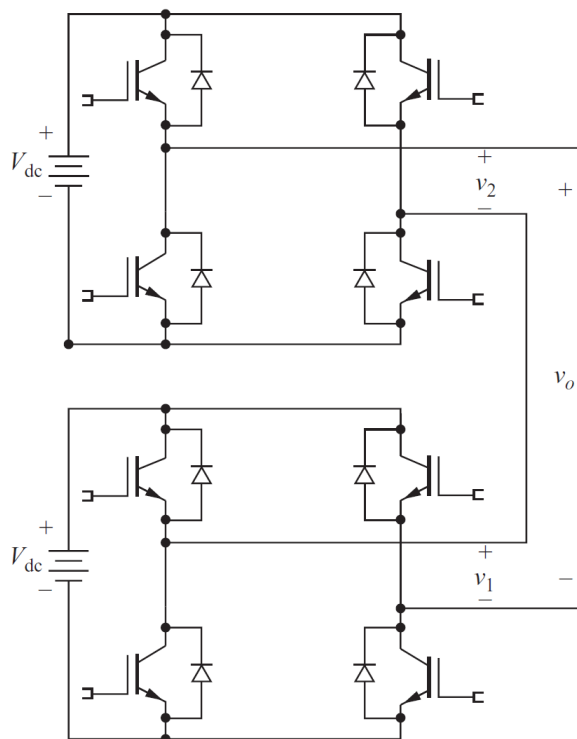
✓ **Select delay angles α_1 and α_2 such that:**

$$\cos(m\alpha_1) + \cos(m\alpha_2) = 0$$

Multi-level Inverters

❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

❖ **With 2 Independent Sources:**



➤ **Amplitude of the fundamental frequency component**



$$V_1 = \left(\frac{4V_{dc}}{\pi} \right) [\cos(\alpha_1) + \cos(\alpha_2)]$$

➤ **Modulation index** is defined as the ratio of the amplitude of the fundamental frequency component of v_o to the amplitude of the fundamental frequency component of a square wave output with input $2V_{dc}$

➤ **To meet the modulation index requirement and eliminate harmonic:**

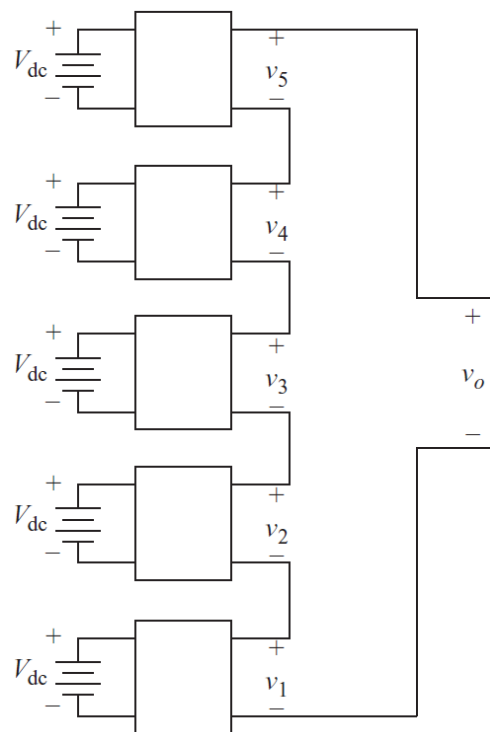
✓ **Solve:**
$$\begin{cases} \cos(\alpha_1) + \cos(\alpha_2) = 2M_i \\ \cos(m\alpha_1) + \cos(m\alpha_2) = 0 \end{cases}$$

$$M_i = \frac{V_1}{4(2 \times V_{dc})} = \frac{\cos(\alpha_1) + \cos(\alpha_2)}{2}$$

Multi-level Inverters

❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

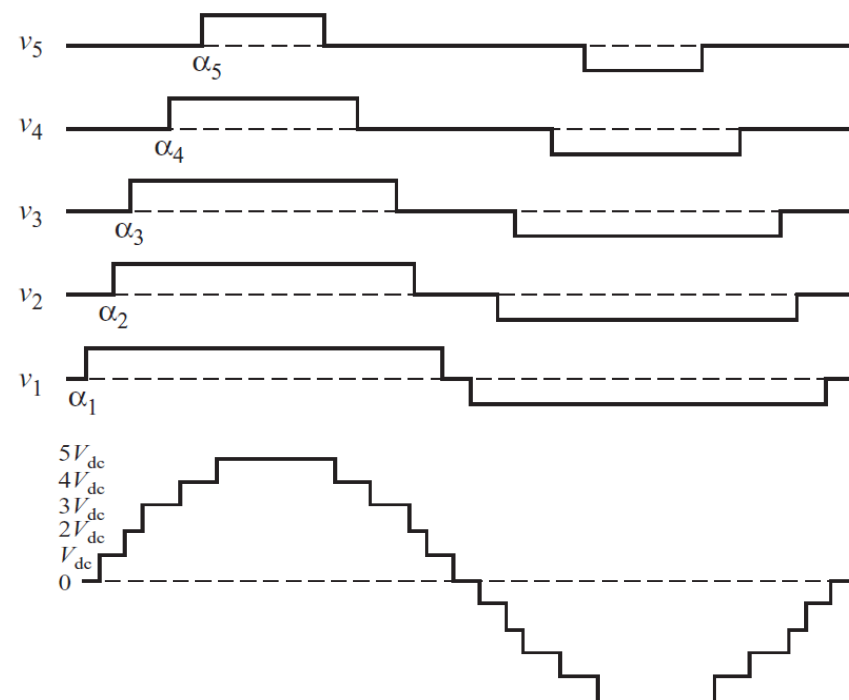
❖ **With k Independent Sources:**



$$v_k(t) = \sum_{n \text{ odd}} V_{n,k} \sin(n\omega_0 t)$$

$$V_{n,k} = \left(\frac{4V_{dc}}{n\pi} \right) \cos(n\alpha_k)$$

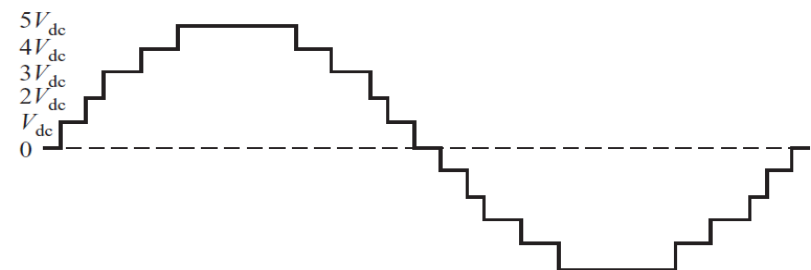
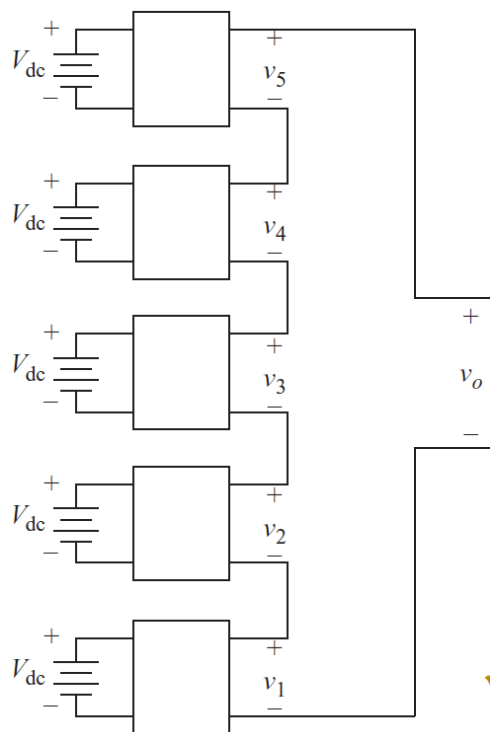
- Each H-bridge can output $+V_{dc}$, 0 or $-V_{dc}$
- The total output voltage can be any combination of these individual output levels
- How many possible output levels for k -source inverter?



Multi-level Inverters

❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

❖ **With k Independent Sources:**



➤ **Fourier series of the total output voltage:**

$$v_o(t) = \sum_{n \text{ odd}} V_n \sin(n\omega_0 t)$$

$$V_n = \left(\frac{4V_{dc}}{n\pi} \right) [\cos(n\alpha_1) + \cos(n\alpha_2) + \cdots + \cos(n\alpha_k)]$$

➤ **How to eliminate the m th harmonic?**

✓ **Select delay angles such that:** $\cos(m\alpha_1) + \cos(m\alpha_2) + \cdots + \cos(m\alpha_k) = 0$

➤ **Is there any other condition to meet?**

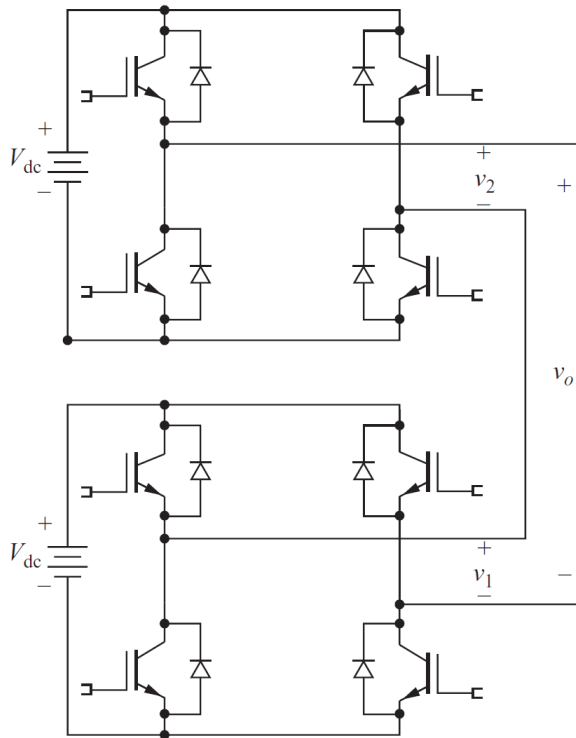
$$M_i = \frac{V_1}{4(kV_{dc})/\pi} = \frac{\cos(\alpha_1) + \cdots + \cos(\alpha_k)}{2}$$

✓ **For k dc sources, $k-1$ harmonics can be eliminated while establishing a particular modulation index**

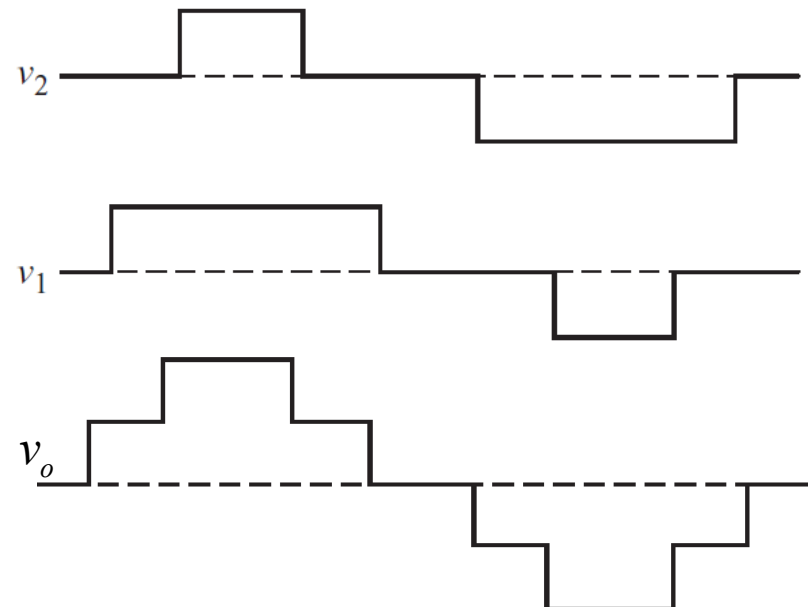
Multi-level Inverters

❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

❖ **Equalizing average source power with pattern/duty swapping:**



- A longer pulse width associated with a source means delivering more power from that source
- If the dc sources are batteries, this means one battery will discharge faster than the other
- We must ensure each dc source conducts (delivers power) for an equal amount of time (on average)

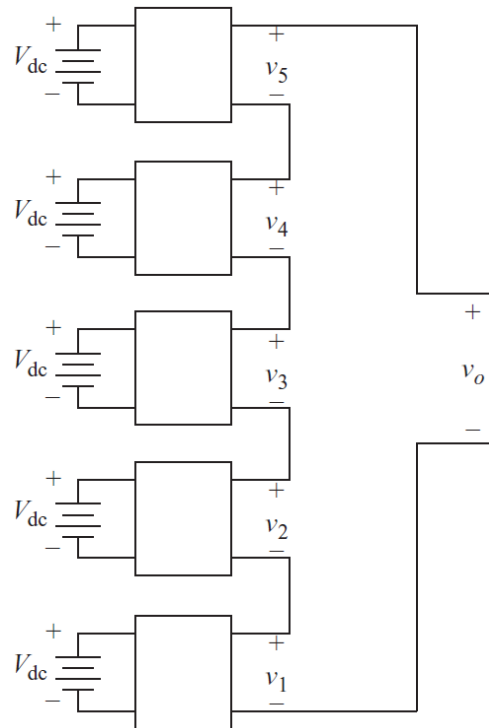


➤ Can this idea be extended to inverters with more levels?

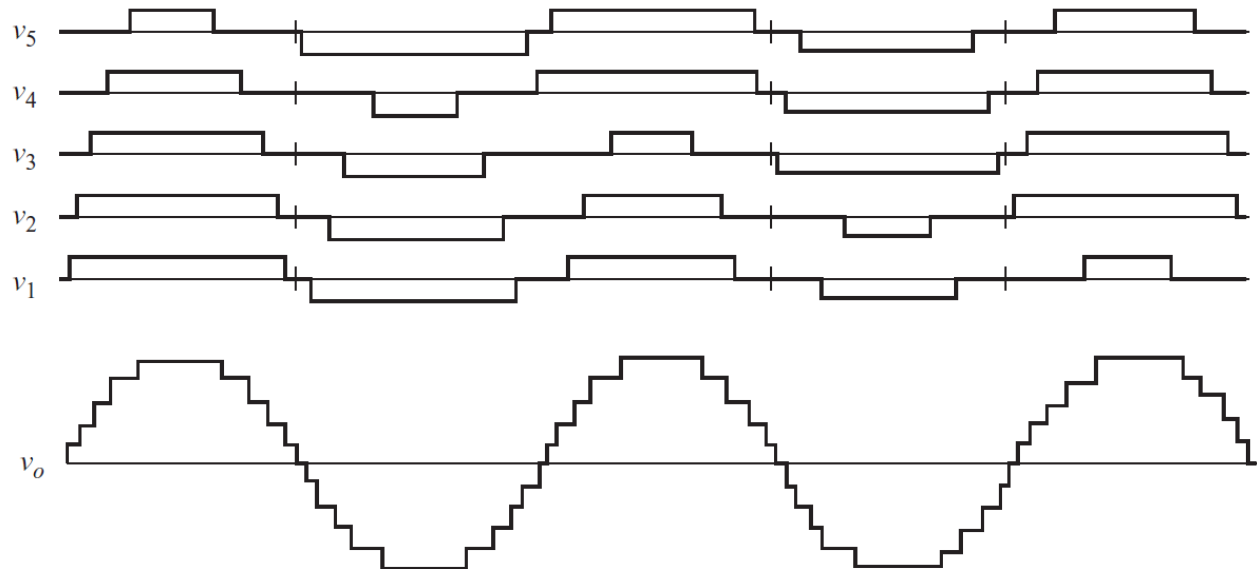
Multi-level Inverters

❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

❖ **Equalizing average source power with pattern/duty swapping:**



➤ We must ensure each dc source conducts (delivers power) for an equal amount of time (on average)



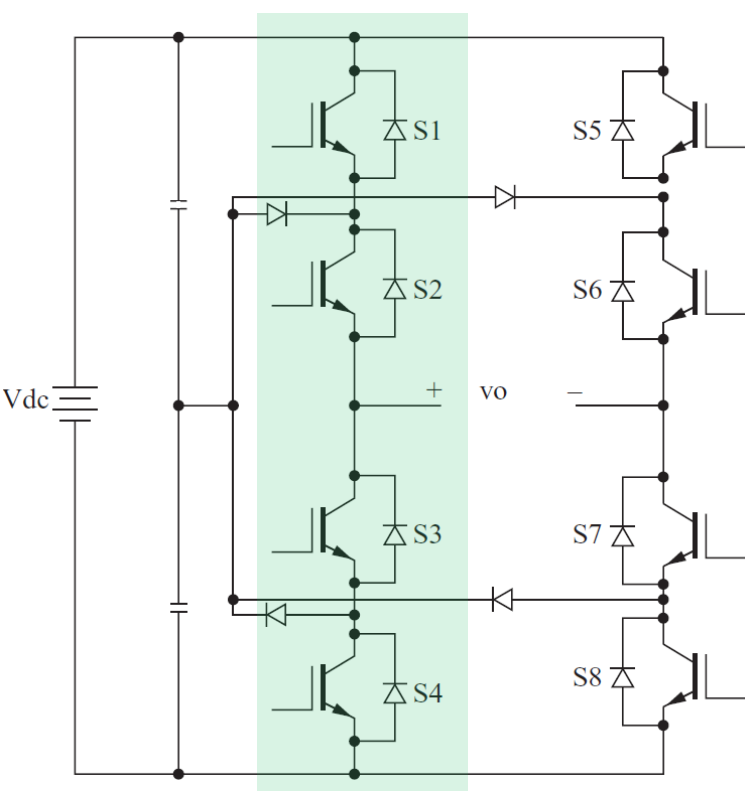
➤ The configurations of multilevel inverters presented so far require multiple independent dc voltage sources. These are best suited in applications in which batteries, fuel cells, or photovoltaics are the dc sources.

Multi-level Inverters

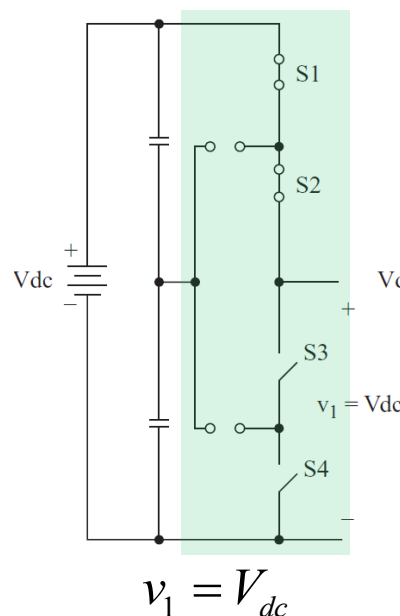
❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

❖ **With Single DC Source:** ➤ The two capacitors are each charged to $V_{dc}/2$

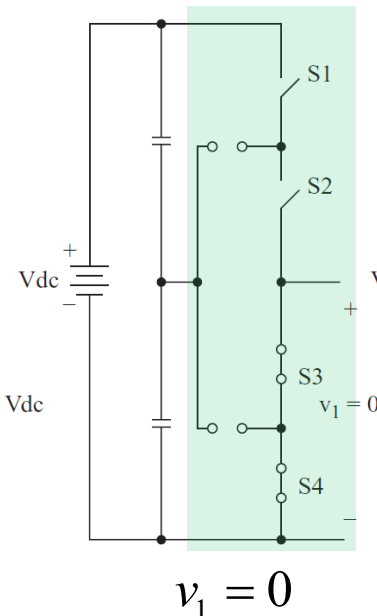
❑ Diode Clamped
multilevel inverter



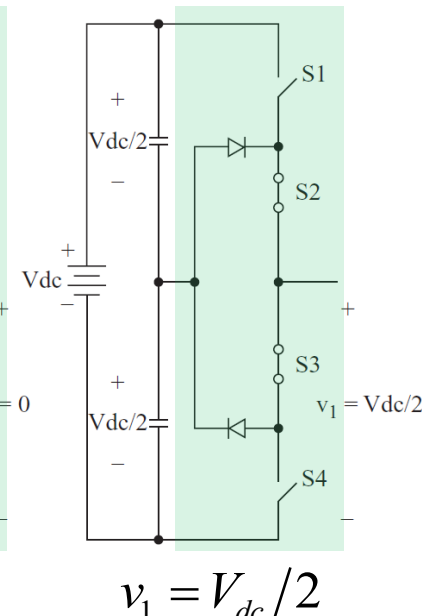
S1 & S2 ON
S3 & S4 OFF



S1 & S2 OFF
S3 & S4 ON



S1 & S4 OFF
S2 & S3 ON



➤ The right-half of the converter can also produce the same voltage levels V_{dc} , 0 and $V_{dc}/2$.

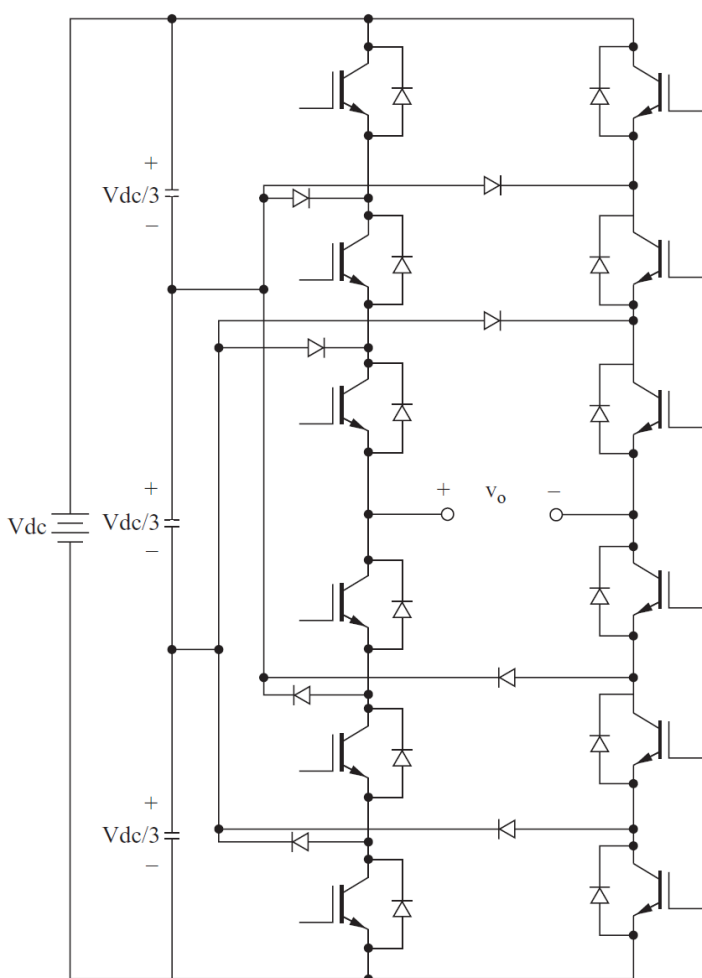
➤ The output voltage can have the levels of V_{dc} , $V_{dc}/2$, 0, $-V_{dc}/2$ and $-V_{dc}$.

Multi-level Inverters

❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

❖ **With Single DC Source:**

❑ Diode Clamped multilevel inverter



➤ The three capacitors are each charged to $V_{dc}/3$

➤ Each half of the converter can produce four voltage levels V_{dc} , $2V_{dc}/3$, $V_{dc}/3$ and 0.



➤ The output voltage can have seven levels of V_{dc} , $2V_{dc}/3$, $V_{dc}/3$, 0, $-V_{dc}/3$, $-2V_{dc}/3$, $-V_{dc}$

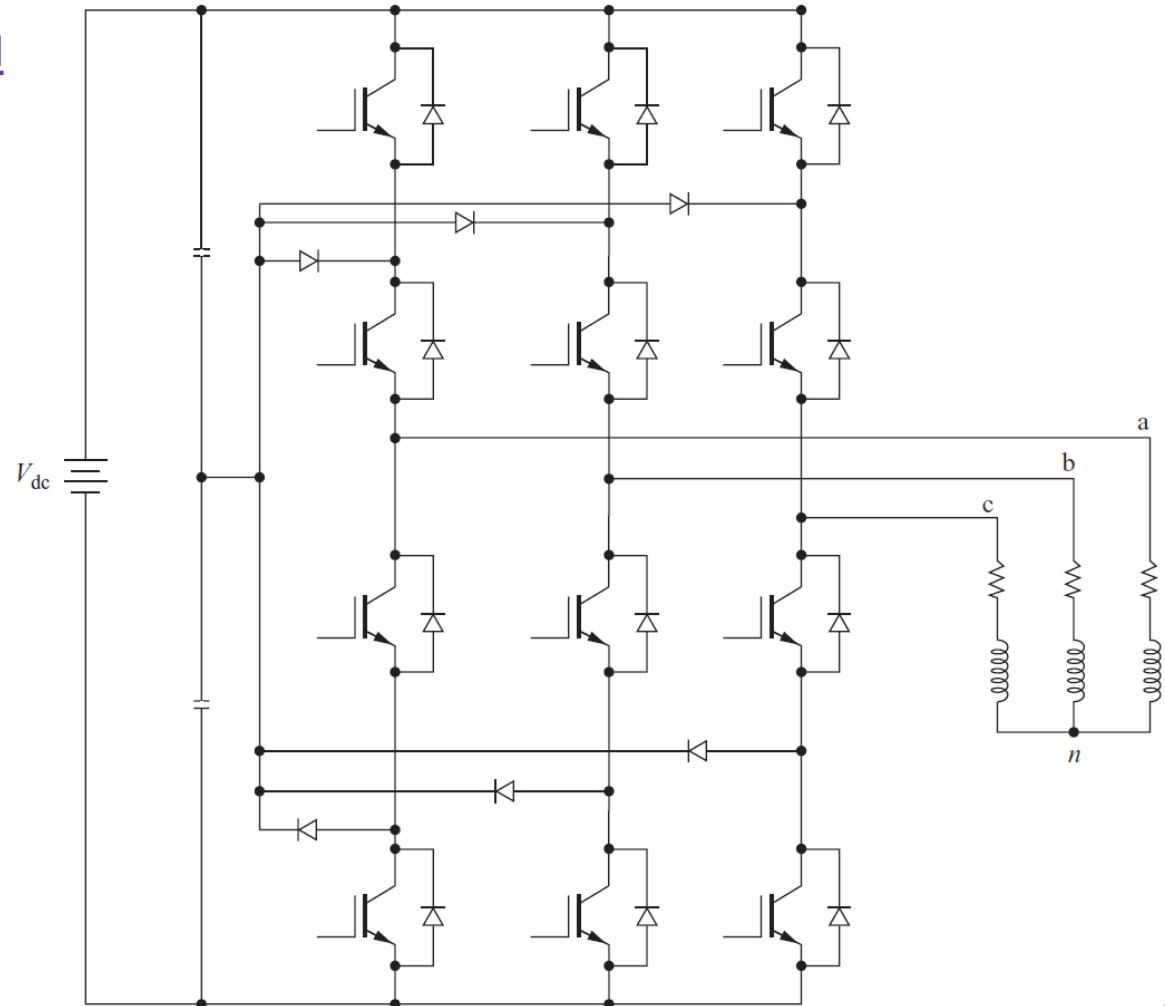
➤ Can we extend this idea to obtain a three-phase output instead of single-phase?

Multi-level Inverters

❑ **Objective:** Producing output voltage levels additional to 0 and $\pm V_{dc}$

❖ **With Single DC Source:**

❑ **Three-Phase Diode Clamped**
multilevel inverter



Space Vector Modulation (SVM)

To be continued...